

Use it or Slowly Lose it: Expertise Atrophy with Organizational AI Usage*

James Siderius[†]

Robert A. Shumsky[‡]

Alva Taylor[§]

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Abstract

Problem definition: As organizations adopt generative AI, its use can improve productivity, but reliance can lead to atrophy of worker knowledge and skills. The challenge is how to incentivize human oversight and sustain long-run expertise. This is difficult because improving AI quality makes oversight and skill maintenance harder to motivate, as acceptable outcomes increasingly occur even when workers disengage.

Methodology/results: Using a principal-agent framework, we study optimal incentive design when workers exert costly effort to verify imperfect AI output, where effort both improves current performance and preserves expertise. Three implications arise. First, profit-maximizing compensation can be non-monotonic in AI quality or skill, and firms may even be more profitable with worse AI systems. Second, due to contracting frictions, firms may allow expertise to deteriorate by substituting effort for expertise, leading to significant performance losses relative to a benchmark where both are prescribed. Third, when tasks are relatively less complex with short learning curves and high returns on effort for low-skilled workers, then the risk of skill atrophy can mitigate these frictions. For such tasks workers are self-motivated to preserve expertise, so higher rates of potential skill loss can, counterintuitively, increase profit.

Managerial implications: We identify when strong incentives are needed to sustain engagement and maintain expertise, and when contracting frictions create tension in providing them. In doing so, we call out potential AI “danger zones.” We discover that moderate-to-high AI performance requires stronger incentives for vigilance. When skill decay is low to moderate, expertise losses may be overlooked yet generate substantial long-term costs, underscoring when proactive monitoring and investment in human capital are most valuable.

Keywords: artificial intelligence, organizational learning, organizational capabilities, human-computer interaction, principal-agent models, skill development

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[†]Tuck School of Business at Dartmouth College, james.siderius@tuck.dartmouth.edu

[‡]Tuck School of Business at Dartmouth College, shumsky@dartmouth.edu

[§]Tuck School of Business at Dartmouth College, alva.h.taylor@tuck.dartmouth.edu

1 Introduction

Rapid advances in generative AI are transforming organizational processes in almost every knowledge domain. Although these systems greatly improve baseline accuracy, they remain imperfect. Even state-of-the-art models exhibit persistent hallucination rates, often exceeding 50% for complex prompts in systems like OpenAI's o3 (OpenAI, 2025), and newer frontier models still display systematic failure modes (Tow Center for Digital Journalism, 2025). As Mollick (2025) observes, AI today is “superhuman in enough areas to change how we work and live, but unreliable enough that human expertise is needed to figure out when AI works and when it doesn't.”

Because of these residual errors, organizations must keep humans “in the loop” to monitor, verify, and correct AI-generated output. Yet oversight is itself an operational challenge. When AI performs well, workers tasked with monitoring its output may reduce vigilance, making errors more likely to slip through. Examples include radiology AI not detecting breast cancers (Woo et al., 2025), content moderation failures on social media (Tadesse, 2025), major newspapers publishing hallucinated reading lists (Blair, 2025) and lawyers citing fictional cases generated by AI (Merken, 2025). These examples underscore how fragile oversight becomes when humans disengage.

Over time, disengagement creates the potential for quiet atrophy, where system outputs appear stable but the human expertise needed to detect and correct errors gradually erodes (Argote, Lee, and Park, 2021). The use of AI may slow the acquisition of basic expertise, lead to the deterioration of existing skills, and prevent experts from acquiring the new knowledge needed to maintain proficiency. For example, Budzyn et al. (2025) finds that the use of AI by endoscopists reduces their ability to detect pre-cancerous growths from a colonoscopy. Natali et al. (2025) reviews the growing evidence that the use of AI by health care providers can lead to a deterioration of skills in physical examination, communication with patients, diagnostic accuracy, and clinical reasoning. In a field experiment with chess club students as subjects, Poulidis, Bastani, and Bastani (2025) find that when students may access AI advice at any point over a series of games, then those students have a lower rate of skill improvement than students for whom AI advice is more limited. Shen and Tamkin (2026) finds that software developers who rely on AI for coding are less likely to obtain new skills, and they comment “...as companies transition to more AI code writing with human supervision, humans may not possess the necessary skills to validate and debug AI-written code if their skill formation was inhibited by using AI in the first place” (p. 19).

The link between AI and the stock of underlying expertise is both a novel and familiar challenge. New technologies have always shifted the stock of expertise, from the printing press (scribing, copying and memorization) to calculators (mental and manual calculations) to GPS (geographic and directional sense). The advance of AI tools similarly shifts expertise needs, yet it is a dramatically different phenomena. While past tools are task specific, generative AI is a generic knowledge tool that can be applied to almost any problem, requiring limited input knowledge to generate an answer (whether correct or incorrect). Leaders often treat AI the way a professor would treat a tireless, almost costless, graduate student: “Have it do the grunt work so that I can focus on higher level questions and issues.” But with a graduate student, every task is also an investment in the next generation of human, critical expertise. With AI, the work gets done, but no one learns. As the system improves, and as more work

is delegated to AI, humans see fewer hard problems and have fewer opportunities for reasoning. Over time the AI may become more reliable but the humans get rustier. At least in the near term, as AI is applied to novel applications with unpredictable results (the ‘jagged frontier’), we will continue to rely on human expertise to ensure the quality of AI output.

Despite the practical importance of these issues, we lack a theory of how firms should structure oversight incentives in AI-enabled systems. Traditional principal-agent models allow effort to respond to incentives and incorporate technology through a production function, but typically treat oversight effort as affecting current-period output only. In contrast, oversight in AI-enabled systems shapes the evolution of human expertise itself. We endogenize the agent’s skill through their reliance on AI and show how rising AI reliability alters opportunities to maintain expertise. As a consequence, the results of our model provide guidance on when firms should pay for diligence, how oversight incentives should change as AI improves, and how these choices shape long-run organizational capability.

This paper develops a model of human-AI co-production in which workers verify and correct AI output and exert costly diligence. Effort improves current accuracy and slows the decay of expertise through potential learning opportunities and fewer instances of forgetting. Workers in our model are compensated based on final output quality in each period (i.e., they are paid a “piece rate”). We analyze both a static-skill environment (skills fixed) and a dynamic-skill environment (skills evolve with effort), showing how AI improvement reshapes incentives, effort, and the distribution of expertise.

We will see that the impact of possible skill atrophy depends upon the task complexity. We define *high-complexity* tasks as tasks for which expertise must be developed over long periods through training or experience, and can be lost more quickly as the field advances. In addition, we assume that for high-complexity tasks, skill and effort are complements, as extra effort by a novice will do little to improve performance, while effort is disproportionately productive in the hands of an expert. We define *low-complexity tasks* as those for which learning is fast and skill decay is slow, and a low-skilled worker can compensate through diligence.

Our analysis yields four main findings:

- (i) We show that oversight incentives follow a three-regime structure — no bonus leading to minimal effort, pay for spot checks (i.e., partial effort), or payment that incentivizes max effort — and that the boundaries between these regimes shift non-monotonically as AI improves. Contrary to conventional wisdom, intermediate AI reliability is the most dangerous zone: effort becomes less pivotal technologically, but also more tempting to shirk, requiring stronger incentives than when AI is highly unreliable or highly reliable. Thus, incremental improvements in AI can increase oversight costs and induce a “critical region” where complacency becomes most severe.
- (ii) When skills evolve over time, the first-best benchmark is simple: absent contracting frictions, it is optimal for workers to exert effort both to improve current AI output and to maintain expertise for the future, regardless of the level of AI reliability. With contracting frictions, however, sustaining expertise becomes costly because maintaining skill requires providing incentives for continued effort. As a result, it can be optimal for the firm to tolerate skill degradation and allow workers to drift toward low skill levels. Nevertheless, relative to a setting in which worker skills are fixed, the

possibility of expertise atrophy raises the stakes of effort: the firm tends to offer higher compensation and induce greater effort to prevent skill loss.

- (iii) The impact of possible skill atrophy depends upon the task complexity. For high-complexity tasks, in equilibrium the workforce is either all-expert when output value is high, or all non-expert for lower output value. In addition, either faster learning or slower decay of skills (weakly) increases the chance of an all-expert workforce and (weakly) increases profits.
- (iv) For low-complexity tasks, there exists a mixed equilibrium with both low and high-skilled workers. In addition, in this environment an organization may attempt to speed up learning or inhibit skill decay, but surprisingly we find that this can be harmful to the firm's results. If expertise degrades slowly or if expertise can be quickly regained, workers are less worried about maintaining skills, so require higher compensation to not fall back on the AI. Therefore, for low-complexity tasks, reducing the learning rate or increasing the rate of skill decay can lead to a higher proportion of high-skilled workers in equilibrium as well as higher profits.

Our results highlight a structural organizational risk: AI is destabilizing not when it is unreliable or perfect, but when it is just reliable enough to induce human disengagement. In this regime, firms may need to supplement incentive pay with direct monitoring of effort and skill, such as managerial oversight or periodic testing. This need arises because the agency problem creates a large and economically meaningful wedge between first-best and implementable outcomes. Overall, the model clarifies how oversight incentives should evolve with AI quality, when human expertise is worth protecting, and when automation can substitute for it.

Related Literature. We build on classic models of output-based incentives and informativeness (Holmström, 1979), linear contracts and multitasking (Holmstrom and Milgrom, 1991, 1994), and relative performance evaluation (Lazear and Rosen, 1981). These foundational models treat the production technology as fixed and focus on how optimal contracts respond to signal precision, incentive intensity, or task substitution. In contrast, our setting introduces an AI baseline whose reliability varies and thereby endogenizes the marginal value of worker diligence. Moreover, we allow effort to interact with worker skill either as substitutes, where diligence compensates for limited expertise, or as complements, where effort is disproportionately productive in expert hands.

More recent research has examined AI/human collaboration in a principal-agent context. Zhang, Dou, and Wu (2025) describe revenue-sharing contracts with content creators who may delegate some tasks to AI. As in our baseline static model, they find that higher AI capability leads to the need for stronger incentives and a reduction in firm profit. Bastani and Cachon (2025) study a model in which a principal offers an output-based contract to an agent who collaborates with an unreliable AI. While in our model the agent may choose among a continuum of effort levels in improving AI output, a key element of their model is a fixed cost for the agent to inspect the AI output. As a result as the AI improves, it becomes dramatically more expensive for the principal to prevent shirking by the agent, and it may be optimal for the principal to block the use of productivity-enhancing AI. Dai and Taylor (2025) study the organizational design of enterprise AI systems, analyzing how firms should set the AI model's temperature (which governs randomness in output) and structure human oversight when

employees interact with generative AI that may produce erroneous responses. Neither [Zhang, Dou, and Wu \(2025\)](#), [Bastani and Cachon \(2025\)](#), nor [Dai and Taylor \(2025\)](#) examine the impact of AI on learning, while our work focuses on the evolution of worker expertise over time.

Our dynamic setting connects to models of skill formation and human-capital accumulation ([Ben-Porath, 1967](#)), as well as career dynamics with learning about ability and assignment ([Gibbons and Waldman, 1999](#)). This research focuses on how workers invest in human capital and how organizations allocate workers over time. Our departure is that effort plays two roles in AI-enabled systems: it raises the accuracy of today’s output and also helps workers retain the expertise needed to oversee the AI in the future. Because diligence both fixes current errors and prevents skills from deteriorating, incentive design must account for how today’s effort choices shape tomorrow’s capabilities. This dual role creates interactions that generate non-monotone responses to learning and skill decay: how quickly workers learn or lose expertise can shift the organization between very different long-run outcomes. This contrasts with standard human-capital models, where production effort affects current productivity while future capability is determined by a separate skill-investment decision. In our setting, the same oversight effort simultaneously determines current output and the evolution of worker skill, so the incentive problem directly governs both short-run performance and long-run capability. In related work [Xu et al. \(2025\)](#) examine the impact of AI on worker knowledge requirements and organizational structure. They formulate a model in which workers may augment their output with AI while managers check the results. They find that higher AI capabilities may lead to “deskilling,” a reduction in the optimal baseline level of worker knowledge. This is different from our model’s concept of deskilling, or atrophy, which is the loss of expertise by the *same workers* over time due to an overreliance on AI.

A growing literature examines how machine predictions interact with human decision-makers. Prior work shows that more accurate or more transparent AI recommendations can reduce managerial effort and lead to strategic under-engagement ([Lu and Tomlin, 2025](#)), that human adjustments can outperform raw algorithmic forecasts when humans correct predictable errors ([Ibrahim, Kim, and Tong, 2021](#)), and that vigilance falls as AI systems improve ([Dell’Acqua, 2022](#)). Related studies identify verification bias ([De Véricourt and Gurkan, 2023](#)) and attention-allocation distortions under rational inattention when AI tools change the distribution of workload ([Boyacı, Canyakmaz, and De Véricourt, 2024](#)). Our paper complements this line of work by focusing on contract design rather than information design: we ask how compensation influences whether and when humans intervene, how these choices evolve with AI reliability, and how they shape the long-run expertise of the organization.

Finally, our framework speaks to empirical findings that AI tools disproportionately benefit lower-skill workers and compress skill differentials ([Brynjolfsson, Li, and Raymond, 2025](#); [Ni et al., 2024](#)). We provide a micro-incentive mechanism behind these observations: improvements in AI reduce the marginal value of expert diligence and simultaneously make it cheaper for non-experts to achieve near-expert baseline performance, thereby compressing the equilibrium distribution of skill-dependent pay and effort.

Paper Outline. This paper is organized as follows. Section 2 presents our co-production model between AI and a human who validates and corrects (if needed) the AI output. Section 3 studies the

organization’s problem under an important benchmark where skills are static over time and do not vary based on AI reliance. Section 4 endogenizes skill evolution as a function of diligence in verifying and correcting the AI. Section 5 concludes; all proofs are presented in Appendix A and our baseline parameter calibration is explained in Appendix B.

2 Model

Consider an environment in which a human collaborates with an AI to produce output that can be assessed as correct or incorrect. Because the AI does not completely automate the task, but rather augments human effort (Brynjolfsson, Chandar, and Chen, 2025), the human can choose a level of effort in working with the AI to verify the joint output and to correct the output if it is wrong. Specifically, at the lowest level of human effort and when working with a human with the lowest-level of expertise, the AI produces correct output with probability p . That is, the AI is inaccurate or hallucinates – and a non-expert human at minimum effort doesn’t catch and correct the problem – with probability $1 - p$.

In period t , a worker has a skill level $s_t \in [0, 1]$ indicating to what extent she is an expert in the task herself (without AI). The worker chooses to exert effort $x_t \in [0, 1]$ to verify and correct (if necessary) the AI with cost $\frac{1}{2}cx_t^2$. Implicitly, we assume worker effort is bounded due to physical limitations, similar to Bimpikis, Ehsani, and Mostaghir (2019), Bond and Gomes (2009), and Schankerman and Schuett (2022). The firm is concerned about the output of the AI/worker pair, which can be low-value or high-value, $v_t \in \{0, h\}$ where $h > 0$. Whether the worker catches and corrects the AI’s mistake is an increasing function in both her skill and her effort levels, i.e., the output is high ($v_t = h$) with probability:

$$\pi_t(s_t, x_t) = 1 - (1 - p) [(1 - \tau)(1 - \theta s_t)(1 - qx_t) + \tau(1 - \theta qs_t x_t)]. \quad (1)$$

We will sometimes refer to $\pi_t(s_t, x_t)$ as the “probability of success.” It is worth noting that “zero effort” should be interpreted as “minimal effort” in our model, because there can still be a boost in productivity even when $x_t = 0$, just from the worker’s skill alone.

The parameter τ captures the degree of complementarity between the worker’s skill and effort. When $\tau = 0$, skill and effort are pure substitutes, so a worker can compensate for low skill with high effort and vice-versa. This would be the case when tasks are relatively simple. A novice can put in extra effort to complete the task, perhaps spending time to obtain the additional knowledge necessary to complete the task (and thus perhaps learning along the way). Experts are already effective at a low level of effort, and the marginal impact of expert effort is lower than that of the novice. In a study of public accountants who were assigned a tax research task, for example, Cloyd (1997) found that time spent searching a tax rules database could substitute for prior expertise. When $\tau = 1$, on the other hand, skill and effort are pure complements, so there are additional returns to effort for high-skilled individuals. This would be the case for high-complexity tasks, where the necessary expertise is developed through longer, prior periods of training and experience. The task is so difficult that extra effort by a novice will do little to improve performance, while extra effort can enable an expert worker to make significant headway. To take an extreme case, effort and expertise are complements when translating Ancient Sumarian into English: a novice barely exposed to the language will have low returns to effort; an

expert's effort will be effective. Many contexts fall somewhere between the pure substitutes and pure complements cases, which would correspond to $\tau \in (0, 1)$.

Another feature of Equation (1) is that $\pi_t(H, 0) - \pi_t(L, 0)$ is shrinking in τ for $H > L$. Thus, the parameter τ also governs the extent to which expertise must be actively deployed to affect outcomes. In complex (high- τ) environments, expertise must be applied: absent active oversight effort, workers (expert or non-expert) primarily pass through AI output, so baseline performance differences are small. In simpler (low- τ) environments, expertise independently improves outcomes even without active intervention. Thus, variation in τ does not eliminate expertise in the complements case; rather, it captures environments in which knowledge is latent unless exercised.

The parameter θ in Equation (1) captures the return on skill in spotting and correcting errors, whereas q is the return on effort. We assume all primitives lie in $[0, 1]$, in particular $q \in [0, 1]$ so that the success probability $\pi_t(s_t, x_t)$ remains bounded in $[0, 1]$. We focus on $\theta \in (1/2, 1)$, which ensures that high skill corresponds to substantively meaningful ability to detect AI mistakes.

Under our principal-agent framework, the organization can only observe whether output is low or high ($v_t = 0$ or $v_t = h$), and the firm pays α to the worker for producing high output in period t . Workers in our model are subject to limited liability and cannot be “punished” for low-value outputs. Hence, it is without loss of generality to assume that there is no bonus awarded for such an outcome. Workers are compensated based on final output quality in each period via a stationary piece rate α for high-value output and zero otherwise. Therefore, the worker's period payoff is $u_t = \alpha\pi_t(s_t, x_t) - \frac{1}{2}cx_t^2$ and the expected period payoff to the organization is $\Pi_t = (h - \alpha)\pi_t(s_t, x_t)$.

Implicit in this formulation is the assumption that worker-firm contracts are stationary. In dynamic employment relationships, rich history-contingent payment paths are often fragile once one allows for renegotiation. In the relational contracting literature (Baker, Gibbons, and Murphy (2002); Levin (2003)), after any public history of performance, past realizations are sunk and the efficient continuation problem depends only on current marginal surplus. Backward-looking punishments or deferred rewards that distort current-period effort are therefore not typically renegotiation-proof and would tend to be renegotiated away. While we do not explicitly model relational enforcement, this logic provides a discipline on our contract space, allowing us to consider renegotiation-proof, piece-rate compensation schemes (see also Gibbons and Roberts (2013)).

This perspective also aligns closely with compensation practices in operational settings. Empirical evidence from manufacturing, service operations, and sales environments documents the widespread use of stable piece rates and commission systems rather than complex, history-dependent payment paths (e.g., Lazear (2000), Shearer (2004), Prendergast (1999)). Even in environments with dynamic learning and productivity evolution, firms typically rely on contemporaneous performance pay that is transparent and administratively simple, and robust to renegotiation. Consistent with both this theoretical foundation and institutional practice, we model the organization's compensation as a stationary piece rate.

The worker discounts payoffs at rate $\delta \in [0, 1)$ whereas the organization maximizes its average payoff (i.e., it is patient over the long horizon). Equivalently, one can think of the worker's discount factor as $\delta \equiv \frac{1}{1+r}$ for some rate $r > 0$, which is isomorphic to setting $r = \frac{1-\delta}{\delta}$. We assume the

organization does not discount its payoffs over time, but this is without loss of generality as long as the firm discounts less aggressively than the worker, which seems reasonable in most contexts (see [Sannikov \(2008\)](#) and [Gibbons and Roberts \(2015\)](#)).

It is also worth noting that the introduction of an AI (i.e., $p > 0$ versus $p = 0$) improves the low-skill workers' productivity more than that of the high-skill workers, leveling the playing field. This has been empirically documented in work such as [Brynjolfsson, Li, and Raymond \(2025\)](#) and [Ni et al. \(2024\)](#). We can see this by noting that

$$\pi_t(s = H, x_t; p) - \pi_t(s = H, x_t; 0) < \pi_t(s = L, x_t; p) - \pi_t(s = L, x_t; 0),$$

for $H > L$. This aspect of the model is therefore consistent with empirical evidence of how AI has affected productivity in the workplace.

3 Benchmark Equilibrium: Static Skills

We first treat skill as constant so that $s_t = s$ for all t (i.e., skills do not improve or atrophy over time, but are simply an attribute of the worker). The corresponding problem is therefore static and can be solved as if it were one-shot. For this reason, we drop the dependence on t in this section, readopting it for the dynamic case of Section 4. Let $(\cdot)_0^1$ denote the function that clips to 0 and 1, i.e., $\min\{1, \max\{0, \cdot\}\}$, write $\pi_0 \equiv 1 - (1 - p)[(1 - \tau)(1 - \theta s) + \tau]$ (the baseline probability of success with minimal effort), and set $K \equiv (1 - p)q[(1 - \tau)(1 - \theta s) + \tau\theta s]$ (the return on effort). Therefore, the probability of high-quality output is, $\pi(s, x) = \pi_0 + Kx$.

3.1 First-Best

For the first-best analysis we assume that the firm can directly observe and enforce the level of worker effort and compensates the worker directly for the cost of effort. The organization solves the effort allocation problem,

$$\max_{x \in [0, 1]} h\pi(s, x) - \frac{1}{2}cx^2. \quad (2)$$

Proposition 1. *The first-best level of effort is $x^{FB} = (\frac{hK}{c})_0^1 > 0$, with the success probability*

$$\pi^{FB} = p + (1 - p)\theta s(1 - \tau) + Kx^{FB},$$

and organizational profit

$$\Pi^{FB} = h\pi^{FB} - \frac{1}{2}c(x^{FB})^2.$$

Moreover, Π^{FB} is monotonically increasing in p , s , and q .

Proposition 1 provides the first-best benchmark. In the absence of incentive constraints, the firm chooses effort to maximize total surplus. The resulting first-best profit is increasing in AI performance, worker skill, and task verifiability, reflecting that improvements in any of these dimensions are beneficial to the organization, as one might expect without contracting frictions.

3.2 Optimal Contract

Under the principal-agent framework, the firm can only observe output, pays the worker α for high-value output ($v = h$), and 0 for low-value output. Our next result characterizes the optimal contract for the organization to maximize profit.

Theorem 1. *The worker's best response is given by $x^*(\alpha) = (\frac{\alpha K}{c})_0^1$. The firm's optimal compensation is given by*

$$\alpha^* = \begin{cases} 0, & \text{if } h < \underline{h} \\ \frac{h}{2} - \frac{c\pi_0}{2K^2}, & \text{if } \underline{h} < h < \bar{h} \\ \frac{c}{K}, & \text{if } h > \bar{h}, \end{cases}$$

where $\underline{h} = \frac{c\pi_0}{K^2}$ and $\bar{h} = \frac{c\pi_0}{K^2} + \frac{2c}{K}$. The equilibrium probability of success is given by

$$\pi^* = \begin{cases} \pi_0, & \text{if } h < \underline{h} \\ \frac{\pi_0}{2} + \frac{hK^2}{2c}, & \text{if } \underline{h} < h < \bar{h} \\ \pi_0 + K, & \text{if } h > \bar{h}, \end{cases}$$

and the firm's per-period profit is given by

$$\Pi^* = \begin{cases} h\pi_0, & \text{if } h < \underline{h} \\ \frac{h\pi_0}{2} + \frac{h^2K^2}{4c} + \frac{c\pi_0^2}{4K^2}, & \text{if } \underline{h} < h < \bar{h} \\ (h - \frac{c}{K})(\pi_0 + K), & \text{if } h > \bar{h}. \end{cases}$$

Theorem 1 characterizes optimal incentives as a function of baseline accuracy π_0 and marginal effort productivity K , yielding three regions defined by output value h :

- (i) When $h < \underline{h}$, the firm provides no incentives. Baseline success π_0 is high relative to the marginal gain from effort K , so additional effort adds little value. The firm relies on the AI-worker baseline and offers no bonus. Unlike the first-best, where effort is strictly positive, contracting frictions make inducing effort too costly, so workers optimally free-ride and effort is zero.
- (ii) When $\underline{h} < h < \bar{h}$, the firm provides interior incentives. AI output remains imperfect, so worker effort is valuable but not worth inducing fully. The firm sets a positive bonus α^* that balances marginal benefit and cost, leading to partial effort, or what we term *spot checking* (one method for implementing interior effort is for the worker to focus on a subset of problems). Effort remains below the first-best due to incentive constraints.
- (iii) When $h > \bar{h}$, correct outcomes are sufficiently valuable that the firm induces maximum effort. It offers just enough bonus to elicit $x = 1$, corresponding to “pay for max effort”: in high-stakes tasks, even small accuracy improvements justify strong incentives.

We illustrate Theorem 1 and subsequent results using parameter values motivated by a real-world use of AI, computer-assisted electrocardiogram (ECG) interpretation. In particular, we use the results

of experiments described in [Kashou et al. \(2023\)](#) to illustrate how model parameters can be derived. These experiments enlisted a variety of health care providers at various levels of expertise, to quantify the benefits of computerized ECG interpretation (CEI). The documented benefits included both accuracy and efficiency, e.g., the CEI reduced the time spent by a provider for each ECG interpretation. Appendix B describes how our parameters are motivated by the published experimental data. The resulting parameters are $p = 0.44$, $\tau = 0.32$, $q = 0.59$, $\theta = 0.58$, $c = 3.89$, and $h = 38.9$. For this Section with the static skills model, we will assume that $s = 0.2$. Our purpose here is not to empirically validate our model or claim any generality for these parameters, but instead to derive baseline parameter values that are consistent with observation and can be used to visualize the model results.

The procedure described in Appendix B uses the time required for an interpretation as a measure of worker effort (x), under the assumption that more interpretation time leads to more accuracy, and is costly to the provider. Because [Kashou et al. \(2023\)](#) focused on the absolute impact of CEI and not the relationship between time and accuracy, we propose a simple model of this relationship. In particular, we assume that there is a correlation of 0.25 between time and accuracy. This assumption affects all parameters, but has the largest effect on q , the return on effort. Later we will see how different assumptions can change the system's output and firm profit.

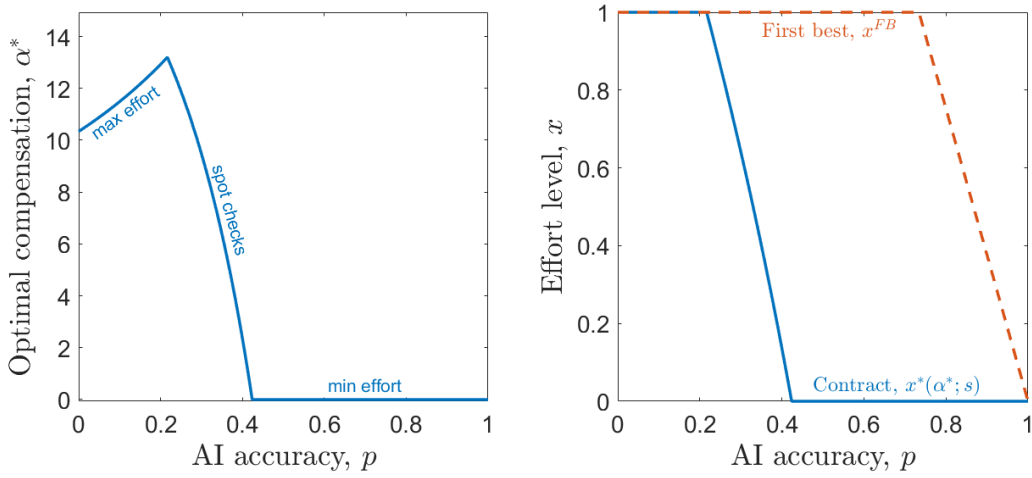


Figure 1. **Left:** Optimal compensation α^* as p varies (using baseline parameters)
Right: Optimal effort $x^*(\alpha^*; s)$ and first-best effort x^{FB} as p varies (using baseline parameters)

Figure 1 shows the three incentive regions implied by Theorem 1, given the baseline parameters. The left panel plots the optimal bonus α^* as AI accuracy p increases. When p is low, human effort is relatively productive, so the firm sets a large bonus to induce full output-checking and correcting (“max effort”). Note that in this region, the optimal compensation rises as p rises (more on this below). As p enters the middle range, the bonus falls sharply: the AI now handles many errors on its own, while the worker becomes increasingly tempted to free-ride on this stronger baseline. Maintaining full effort would require a larger bonus, but the firm benefits less from each additional unit of diligence, so it optimally switches to a regime of partial incentives (“spot checks”). Once p is high enough, the baseline is so strong, and the incentive cost so large, that paying any bonus is wasteful, and α^* collapses to zero (“min effort”).

The right panel shows the corresponding effort levels. The blue (solid) curve is the worker's effort chosen under the optimal contract; the orange (dashed) curve is the first-best effort. Under the optimal contract, effort remains at its maximum only when p is very low. As soon as p enters the middle region, effort declines steeply because the firm no longer finds it worthwhile to pay the larger bonuses required to sustain full diligence. Eventually, once p is sufficiently high, contractual effort drops to zero even though the first-best would still call for positive “last-mile” checking until the AI becomes nearly perfect. The gap between the two curves illustrates the core contracting friction: long before effort becomes useless technologically, it becomes too expensive to motivate.

3.3 Compensation Levels

While Theorem 1 establishes the three possible incentive regimes, it does not describe how the optimal compensation responds to changes in the underlying primitives. In particular, managers may want to know whether changes in AI accuracy, worker skill, or return on effort will reliably increase or decrease the strength of incentives. The next result shows that rather than being monotone, optimal pay is *single-peaked* in each of these parameters (as we saw in Figure 1, Left panel). This captures the intuition that incentives are strongest when AI and worker skill are of intermediate quality — good enough that human checks matter, but not so good that they make extra effort redundant.

We provide a definition that will be useful for the following result.

Definition 1. A property holds *generically* if there exists an open set $\mathcal{I}$ of primitives such that the property holds over all of $\mathcal{I}$.

Proposition 2.

- (a) The compensation α^* is generically single-peaked in p (first increasing, then decreasing); moreover, if $p^*(\tau)$ is the p that maximizes α^* , then there exists $0 < \underline{s}^* < \bar{s}^* < 1$ such that $p^*(\tau)$ is increasing in τ when $s > \bar{s}^*$ but decreasing in τ when $s < \underline{s}^*$;
- (b) The compensation α^* is generically single-peaked in q (first increasing, then decreasing); moreover, if $q^*(\tau)$ is the q that maximizes α^* , then there exists $0 < \underline{s}^* < \bar{s}^* < 1$ such that $q^*(\tau)$ is decreasing in τ when $s > \bar{s}^*$ but increasing in τ when $s < \underline{s}^*$.
- (c) The compensation α^* is generically single-peaked in s if $\tau > 1/2$, otherwise it is generically single-troughed (U-shaped) in s if $\tau < 1/2$.

Proposition 2 shows that compensation is often non-monotonic as AI accuracy p , return on effort q , and worker skill s increase. The core force is straightforward: as AI improves, baseline performance rises, making workers more tempted to free-ride while reducing the incremental value of correcting remaining errors. As a result, incentives are weak when AI is either very poor (effort has little impact) or very strong (little left to fix), and strongest at intermediate levels of AI accuracy, producing a three-phase pattern of no pay, pay for spot-checks, and pay for max effort.

Return on effort q generates a similar hump. When this return is low, effort is ineffective and bonuses are low; when problems are moderately tractable, effort meaningfully improves outcomes

and incentives rise; and when problems become very easy, additional oversight adds little value and incentives decline.

Worker skill s interacts with effort depending on whether the two are complements or substitutes. If they are complements, higher skill leads to greater returns, so incentives initially rise with skill before falling once baseline performance becomes strong enough that oversight adds little value. If they are substitutes, greater skill reduces the need to incentivize effort because skilled workers already correct many errors; however, at very high skill levels, firms may again pay strongly to activate rare but valuable expertise. In short, AI can compress incentives in the middle of the distribution while preserving strong incentives for the highest-skill experts, with the exact pattern determined by whether skill amplifies or replaces effort.

3.4 Organizational Profit

Our next result characterizes how firm profits change, as a result of the three parameters (p , s , and q) that we studied in Proposition 2.

Proposition 3. *Firm profit Π^* is:*

- (a) *Monotonically increasing in q ;*
- (b) *Generically non-monotone in p ;*
- (c) *Generically non-monotone in s when $\tau < 1/2$, and increasing in s when $\tau > 1/2$.*

Part (a) of Proposition 3 shows that greater return on effort q raises profits whenever incentives are active. Once the firm uses incentives, a higher q makes each unit of effort more effective and lowers the bonus needed to induce it, so profits rise. If incentives are not used ($\alpha^* = 0$), q is irrelevant; once they are, profits increase monotonically in q .

Parts (b) and (c) explain why profits can be non-monotone in AI quality p and skill s . Higher p or s improves baseline performance, but it also strengthens the incentive problem: because good outcomes become more likely even when workers shirk, workers are more tempted to free-ride on the AI or their own baseline expertise. To prevent free-riding and sustain effort, the firm must offer higher compensation, which can be expensive. When effort is initially valuable (e.g., when effort costs are low), the firm wants to keep workers engaged; as p or s rise, doing so requires a rapidly increasing bonus, and the added wage bill can outweigh the direct benefit of better baseline accuracy, causing profits to fall over an intermediate range. When effort is costly, the firm relies less on incentives to begin with, so this “anti-free-riding” compensation channel is weaker and profits are more likely to rise with p and s .

With substitutes (low τ), increases in skill make effort less pivotal, so preventing free-riding becomes more expensive relative to the benefit of additional effort, making non-monotonicity in s more likely. With complements (high τ), skill raises the return to effort, so the firm can justify and more easily sustain incentives, making profits more likely to be monotone in s . AI quality p can still generate a hump when effort is sufficiently cheap, though the peak shifts with τ because complementarity changes how valuable it is to pay for vigilance at higher baseline accuracy.

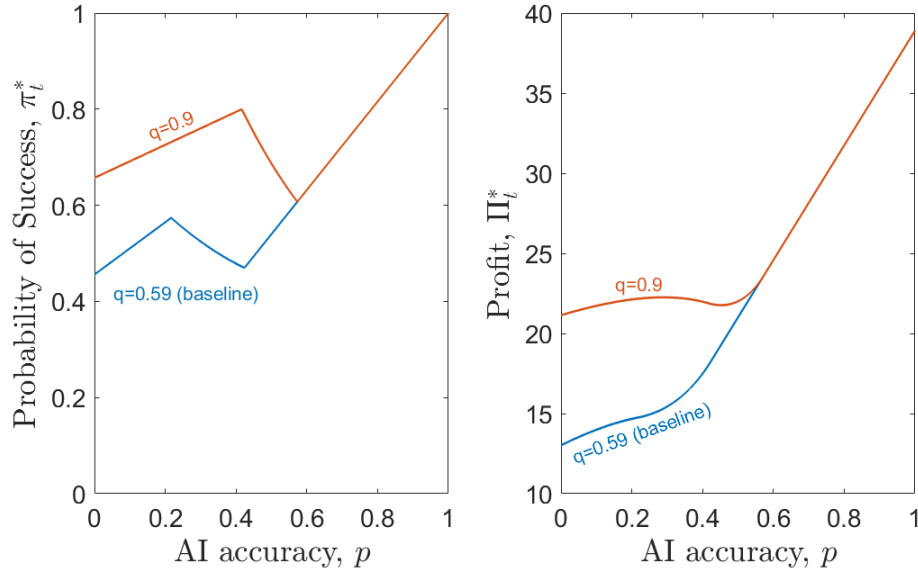


Figure 2. Optimal output and profit as p varies, for two values of q , return on effort. All other parameters are set to their baseline values.

Figure 2 illustrates how these contracting frictions translate into non-monotonic patterns in both success probabilities and profits as AI accuracy improves. For each of the two levels of return on effort, the probability of success initially rises with p , reflecting the higher quality of the AI baseline, but then dips once p enters the region where the firm optimally scales back incentives. This decline occurs precisely because the cost of motivating diligence grows faster than the technological benefit of correcting the AI's remaining errors, leading the firm to induce less effort even though additional checking and correcting remains socially valuable. The resulting drop in success probability produces a corresponding dip in profit in the curve for $q = 0.90$. Importantly, this pattern is entirely a consequence of the incentive problem: in the first best, effort never declines until the AI is nearly perfect, so profit would be strictly increasing in p . The figure therefore highlights a central implication of the model: AI progress can temporarily reduce output and profits in the presence of contracting frictions, even though it would generate smooth performance gains in a frictionless benchmark.

4 Equilibrium in the Dynamic Model

Now, we consider the evolution of skills over time. We assume that worker effort has a short-term error-checking-and-correcting benefit, but also helps to keep the worker's skills fresh and may develop new skills. In particular, we let there be two skill levels: an expert (H) and non-expert (L , with $L < H$). With extensive effort, a non-expert worker can become an expert and with lack of maximal effort an expert can devolve into a non-expert. That is, if $s_t = L$ and x_t is the effort exerted, then,

$$s_{t+1} = \begin{cases} L, & \text{with probability } 1 - \phi_u \mathbf{1}_{x_t=1} \\ H, & \text{with probability } \phi_u \mathbf{1}_{x_t=1}, \end{cases}$$

where $\mathbf{1}_z$ is equal to 1 if condition z is true and is 0 otherwise. Likewise, without full effort an expert worker can drift downwards; if $s_t = H$ and x_t is the effort exerted, then

$$s_{t+1} = \begin{cases} L, & \text{with probability } \phi_d \mathbf{1}_{x_t < 1} \\ H, & \text{with probability } 1 - \phi_d \mathbf{1}_{x_t < 1} \end{cases}$$

Instead of requiring maximum effort $x_t = 1$ to possibly move from L to H and to avoid moving from H to L , we could define a threshold $0 < E \leq 1$, where any effort $x_t \geq E$ would enable upward (and prevent downward) movement. All of the following results hold true for this more general model, at the cost of additional notation. We find that a model with a continuous relationship between effort and learning/forgetting, rather than a threshold, is intractable.

Note that the parameter ϕ_u represents the rate of expertise acquisition by non-experts, or the *learning rate*. The parameter ϕ_d is the rate at which experts may slip due to both the loss of existing knowledge as well as the failure to acquire the new knowledge needed to maintain expert status in an evolving field; we call ϕ_d the skill *decay rate*.

There is an infinite horizon $t = 1, 2, \dots$ and we assume the worker starts as a non-expert ($s_1 = L$). For any given piece-rate α , the worker maximizes her discounted expected payoff

$$\bar{U}(\alpha) = \max_{\{x_t\}} \mathbb{E} \left[\sum_{t=1}^{\infty} \delta^{t-1} u_t(x_t; \alpha) \right],$$

where the expectation is taken over the stochastic skill path induced by the worker's effort choices. The firm chooses α to maximize its long-run average payoff

$$\bar{\Pi} = \max_{\alpha} \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T (h - \alpha) \pi_t(x_t^*(\alpha)),$$

where the dependence of outcomes on the underlying skill process is suppressed in the notation.

Define $\pi_{0,L}$, $\pi_{0,H}$ and K_L , K_H as follows:

$$\begin{aligned} \pi_{0,s} &= 1 - (1-p)[(1-\tau)(1-\theta s) + \tau], \\ K_s &= (1-p)q[(1-\tau)(1-\theta s) + \tau\theta s]. \end{aligned}$$

The timing is as follows. At $t = 1$, the firm commits to a stationary output-contingent piece rate α (see Section 2 for a discussion of this assumption). Thereafter, in each period t , the worker observes her current skill state $s_t \in \{L, H\}$ and chooses effort $x_t \in [0, 1]$. Output realizes, pay is made, and skill transitions occur. We study Markov-perfect equilibria in stationary strategies.

Proposition 4. *Both $\bar{U}(\alpha)$ and $\bar{\Pi}$ are well-defined. Moreover, there exists a unique Markov-perfect equilibrium (excluding knife-edge cases).*

In the following subsection we characterize the optimal allocation of worker effort, and optimal profits, under the assumption that the firm can monitor and enforce effort in each period. Then in

Section 4.2 we examine how the firm may offer optimal incentives, given that the firm offers rewards for output - not effort - and does not enforce the evolution of skills over time. In Sections 4.3 and 4.4 we obtain additional results by focusing on high vs. low-complexity tasks, defined by the relative task learning and decay rates, as well as whether skill and effort are complements or substitutes.

4.1 First-Best

In the first-best stationary effort allocation problem, but now with skill dynamics, the organization's problem involves three cases:

- (i) Exert interior effort for low skill types (all- L steady state):

$$\max_{x_L} h\pi(L, x_L) - \frac{1}{2}cx_L^2.$$

- (ii) Exert full effort for low skill types but interior effort for experts (mixed steady state):

$$\mu_L \left(h\pi(L, 1) - \frac{1}{2}c \right) + \mu_H \left(\max_{x_H} h\pi(H, x_H) - \frac{1}{2}cx_H^2 \right).$$

where $\mu_L = \frac{\phi_d}{\phi_u + \phi_d}$ and $\mu_H = \frac{\phi_u}{\phi_u + \phi_d}$.

- (iii) Exert full effort for both types (all- H steady state):

$$h\pi(H, 1) - \frac{1}{2}c.$$

This yields the following general characterization of the first-best effort in the dynamic setting.

Proposition 5. *There exist $0 < \underline{h}^{FB} \leq \bar{h}^{FB}$ such that:*

- (i) *If $h > \bar{h}^{FB}$, the first-best effort level is $x_L^{FB} = x_H^{FB} = 1$ with steady-state profit $\Pi^{FB} = h(\pi_{0,H} + K_H) - \frac{1}{2}c$;*
- (ii) *If $\underline{h}^{FB} < h < \bar{h}^{FB}$, the first-best effort level is $x_L^{FB} = 1$ and $x_H^{FB} = \frac{hK_H}{c}$ with steady-state profit $\Pi^{FB} = \mu_L \left(h(\pi_{0,L} + K_L) - \frac{1}{2}c \right) + \mu_H \left(h\pi_{0,H} + \frac{h^2K_H^2}{2c} \right)$;*
- (iii) *If $h < \underline{h}^{FB}$, the first-best effort level is $x_L^{FB} = \frac{hK_L}{c}$ with steady-state profit $\Pi^{FB} = h\pi_{0,L} + \frac{h^2K_L^2}{2c}$.*

Moreover, $\bar{\Pi}^{FB}$ is monotonically increasing in ϕ_u and monotonically decreasing in ϕ_d .

The dynamic first-best clarifies how an organization would allocate effort if it could directly control behavior and exploit the long-run benefits of learning without facing incentive constraints. Unlike the static benchmark in Section 3, the organization now internalizes that effort not only improves performance today but also shapes the distribution of worker types tomorrow. As a result, the planner compares the value of three possible steady states: a low-skill steady state in which effort is interior, a mixed steady state in which non-experts exert full effort and experts receive interior allocations, and a high-skill steady state in which both types exert full effort and the system converges to all- H .

Which regime is optimal depends on the stakes h and on the underlying learning and decay parameters (ϕ_u, ϕ_d) .

The left panel of Figure 3 illustrates the three regions described in Proposition 5. The figure is generated using the baseline parameters described in Appendix B. We see in the Figure that in the first best, non-experts exert interior effort only in a narrow region on the left and top of the panel where the output value h is too low to justify pushing them to the frontier. There is also a purple region in the middle, where non-experts and experts mix. In the green region to the right, only experts exist, and they exert maximum effort.

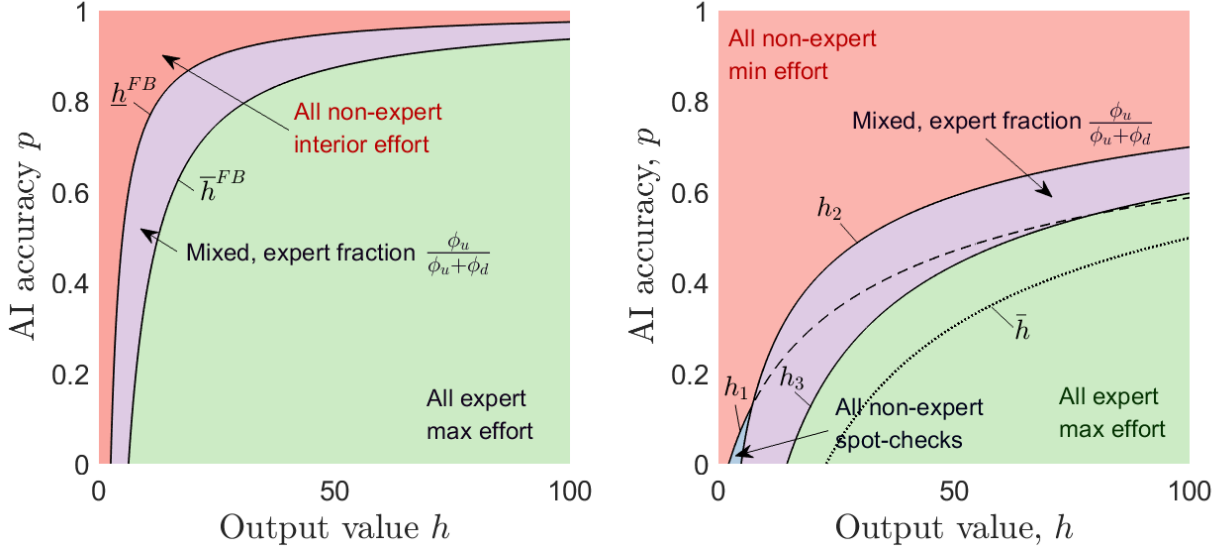


Figure 3. **Left:** First-best skill levels as AI accuracy p and output value h vary at baseline parameters; **Right:** Equilibrium skill levels under contracting as AI accuracy p and output value h at baseline parameters.

This benchmark is important for two reasons. First, it highlights that even without contracting frictions, the optimal allocation of effort is shaped by the organization's ability to build and retain expertise over time. Second, it provides a reference point for understanding the distortions that arise in equilibrium under the optimal contract: while higher ϕ_u or lower ϕ_d always raise value in the first-best, these same parameters may reduce implementable incentives when workers are contracted to voluntarily exert effort. Thus, the dynamic first-best delineates the frontier of what the organization could achieve technologically and serves as the baseline against which the incentive-constrained outcomes in Sections 4.2 to 4.4 are evaluated.

4.2 Optimal Contract

Our next result characterizes the unique steady-state Markov-perfect equilibrium under dynamic skill transitions for the worker.

Theorem 2. *There exist $0 \leq h_1 \leq h_2 \leq h_3$ and $0 < \underline{\alpha} < \bar{\alpha}$ such that:*

- (i) All expert, maximum effort: If $h > h_3$, then $\alpha^* = \bar{\alpha}$, so $x_t^* = 1$ and in steady state the worker is an expert. The steady-state success probability is $\bar{\pi}^* = \pi_{0,H} + K_H$, and the firm's profit is $\bar{\Pi}^* = (h - \alpha_H^*)(\pi_{0,H} + K_H)$;
- (ii) Mixed non-experts at maximum effort with interior experts: If $h_2 < h < h_3$, then $\alpha^* = \frac{h}{2} - \frac{A}{2B} \in [\underline{\alpha}, \bar{\alpha})$ (or $\alpha^* = \underline{\alpha}$ if no such α^* exists), so $x_t^* = 1$ if $s_t = L$ and $x_t^* = \frac{K_H \alpha^*}{c} \in (0, 1)$ if $s_t = H$, and in steady state the worker is a non-expert (resp. expert) with probability $\frac{\phi_d}{\phi_u + \phi_d}$ (resp. $\frac{\phi_u}{\phi_u + \phi_d}$). The steady-state success probability is $\bar{\pi}^* = A + B\alpha^*$, and the firm's profit is $\bar{\Pi}^* = (h - \alpha^*)(A + B\alpha^*)$;
- (iii) All non-expert spot-checks: If $h_1 < h < h_2$, then $\alpha^* = \frac{h}{2} - \frac{c\pi_{0,L}}{2K_L^2} < \underline{\alpha}$, so $x_t^* = \frac{K_L \alpha^*}{c} \in (0, 1)$ when $s_t = L$ and in steady state the worker is a non-expert. The steady-state success probability is $\bar{\pi}^* = \frac{\pi_{0,L}}{2} + \frac{hK_L^2}{2c}$, and the firm's profit is $\bar{\Pi}^* = \left(\frac{h}{2} + \frac{c\pi_{0,L}}{2K_L^2}\right) \left(\frac{\pi_{0,L}}{2} + \frac{hK_L^2}{2c}\right)$;
- (iv) All non-expert, minimum effort: If $h < h_1$, then $\alpha^* = 0$, so $x_t^* = 0$ and in steady state the worker is a non-expert. The steady-state success probability is $\bar{\pi}^* = \pi_{0,L}$, and firm's profit is $\bar{\Pi}^* = h\pi_{0,L}$;

where $A \equiv \frac{\phi_d}{\phi_u + \phi_d}(\pi_{0,L} + K_L) + \frac{\phi_u}{\phi_u + \phi_d}\pi_{0,H}$ and $B \equiv \frac{\phi_u}{\phi_u + \phi_d} \frac{K_H^2}{c}$. Moreover, $h_3 < \bar{h}$ (for whichever skill type binds) so the firm pays for full diligence over a larger set of primitives relative to the optimal contract for the static case described in Theorem 1.

Theorem 2 reveals a rich sequence of possible regimes. For very low stakes ($h < h_1$, region (iv)), the firm does nothing: no effort is elicited and a worker stays at the novice level in steady state. Once h passes the first threshold, the firm begins paying an interior bonus, region (iii), but because $x_L < 1$, promotions never occur and the long-run skill composition still collapses to non-experts only. In region (ii) the firm finds it optimal to incentivize non-experts to put in full effort, guaranteeing promotions can occur. But experts are left at interior effort (sometimes shirking), so the skill distribution in steady state is a non-degenerate mix of non-experts and experts. In other words, the steady state has “diligent non-experts, half-hearted experts,” which runs counter to the naïve intuition that the most skilled workers should be the most tightly incentivized. Finally, once stakes are high enough ($h > h_3$, region (i)), the firm pays sufficiently so that both types are working at max effort, and the system absorbs in the expert state with full diligence. These four regions are illustrated in the right-hand panel of Figure 3, generated using the baseline parameters.

By comparing the left and right panels of Figure 3, we see that the regions under optimal contracting differ significantly from the first best. Under first best, experts exert maximal effort over a wider region. Under contracting, firms want to sustain max effort by experts yet they often cannot, since eliciting that diligence becomes too expensive at intermediate AI accuracy. As a result, the large red region emerges where workers optimally exert minimal effort even though the first best would call for meaningful engagement. This comparison is inverted, however, when we consider the static contracting case of Section 3, delineated by the dotted curve $\bar{h}$ on the lower right. Because in the dynamic case skills can be developed, the benefits of maintaining expertise through max effort hold over a wider region, shifting work toward skilled, high-effort production.

Figure 4 shows the magnitude of contracting distortions, the absolute loss relative to first best (on the left) and the loss as a percentage of first best (on the right) as a function of AI accuracy p and

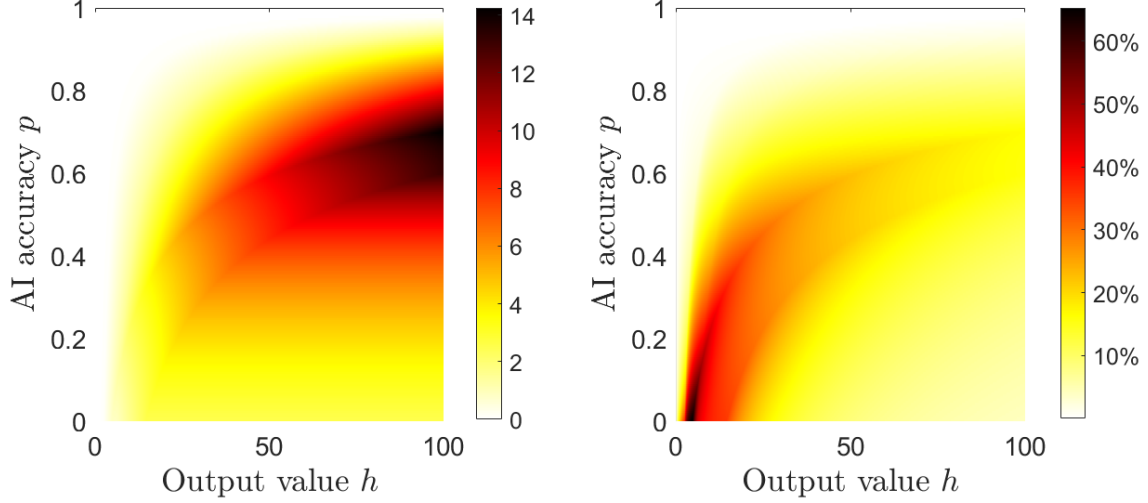


Figure 4. Loss of profit due to agency problem at baseline parameters. **Left:** Absolute (gross) loss, $\bar{\Pi}^{FB} - \bar{\Pi}^*$; **Right:** Percentage loss, $(\bar{\Pi}^{FB} - \bar{\Pi}^*)/\bar{\Pi}^{FB}$

output value h , with other parameters at baseline. The figure reveals two “ridges” of distortions along the borders between the regions shown on the right of Figure 3. For example, just to the left of the ridge on the left the firm suffers because the non-experts are passing along AI output with little oversight, while just to the right of the ridge the firm overpays to incentivize non-experts to work at maximum effort and develop their skills. We can see in this figure that there is an “AI danger zone” at intermediate AI reliability, where incentives are most costly and agency distortions are largest.

Now for ease of exposition, we focus on two distinct cases that capture qualitatively different task environments. First, for *high-complexity* tasks, expertise is necessary for effort to be effective and skills are difficult to acquire but easy to lose. Formally, we assume skill and effort are complements ($\tau > 1/2$) and that demotion risk exceeds promotion likelihood ($\phi_d \geq \phi_u$). In such environments, novice effort alone does little to improve outcomes, and maintaining expert status requires sustained engagement; absent strong incentives, the workforce may drift toward low expertise. In contrast, for *low-complexity* tasks, effort can substitute for skill and expertise is relatively easy to learn and retain. Here we assume skill and effort are substitutes ($\tau < 1/2$) and that promotion is more likely than demotion ($\phi_u \geq \phi_d$). In these settings, lower-skill workers can compensate through effort, and expertise is more easily rebuilt, leading to different incentive and steady-state patterns. Note that our base case parameters of $\tau = 0.32$, $\phi_u = 0.1$, and $\phi_d = 0.05$ are consistent with this low-complexity characterization. While interpreting an ECG is certainly a complex task for someone who is not a medical clinician, our characterization of low vs. high-complexity is driven by the relative rates of learning and decay, and the rate of substitution between skill and effort, rather than the absolute complexity of the task. We will see that these two task-types illustrate how the interaction between technological complementarity and the dynamics of learning and skill decay shapes optimal oversight incentives and long-run organizational capability.

4.3 Contract Structure for High-Complexity Tasks

Let μ_H^* denote the steady-state share of experts in equilibrium. Our next result shows that under high-complexity tasks, an all-or-nothing structure emerges for worker expertise.

Theorem 3. *Let $\tau > 1/2$ and $\phi_u \leq \phi_d$. Then region (ii) of Theorem 2 disappears ($h_2 = h_3$). In particular, $\mu_H^* \in \{0, 1\}$.*

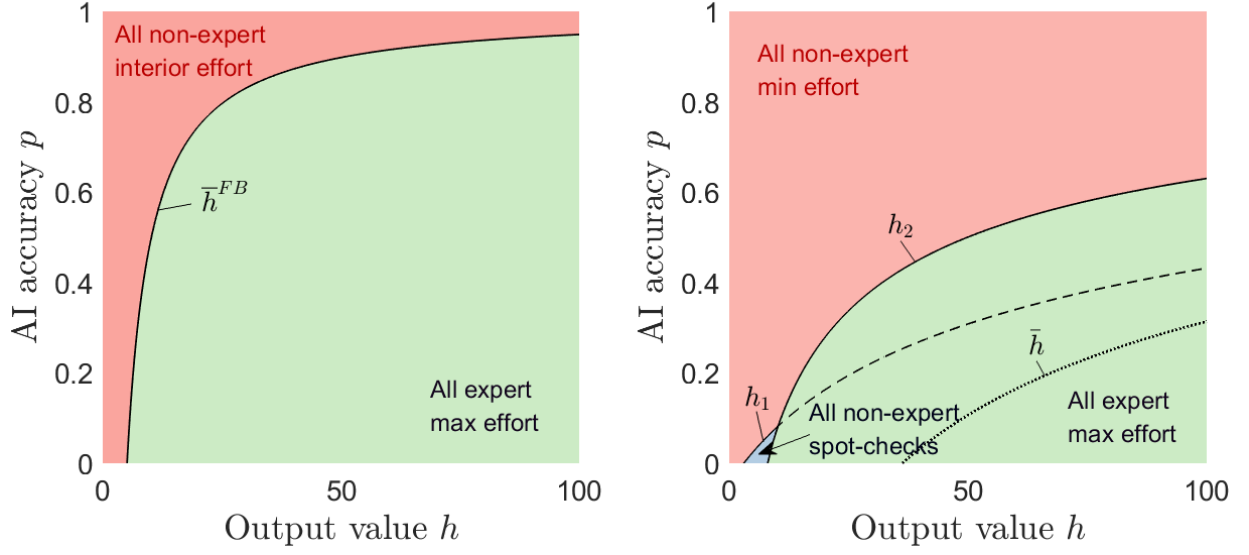


Figure 5. **Left: First-best skill levels as AI accuracy p and output value h vary for a high-complexity task ($\tau = 0.64, \phi_u = 0.05, \phi_d = 0.1$).** **Right: Equilibrium skill levels under contracting as AI accuracy p and output value h for the high-complexity task.**

When tasks have high-complexity, expertise and effort reinforce one another: skilled workers get much more out of careful oversight, and sustained engagement is necessary to remain proficient. At the same time, expertise is difficult to build and can be lost relatively quickly. In such environments, incentives have a stark structure. Figure 5 displays this structure, generated with $\tau = 0.64, \phi_u = 0.05, \phi_d = 0.1$, and all other parameters at baseline. In the panel on the left, the first-best solution allows for a narrow region of all non-experts, and a larger region with only experts. In the right panel, under contracting, there is a large area of non-experts with minimum effort, a small area with spot-checks, and an area in the lower right with all experts. The loss of profit, relative to first-best, is similar to the losses under base case parameters, as in Figure 4, with the largest losses occurring along the boundaries between regions.

The resulting pattern is essentially “all-or-nothing” and there is no regime that mixes non-experts and experts. Firms either invest heavily to keep expertise sharp through persistent monitoring and feedback, or they rely primarily on automation and allow human capability to decay. Partial incentives buy temporary gains but do not preserve long-run skill. Managerially, this implies that complex, high-stakes tasks require a clear strategic choice: either commit to sustained investment in human oversight or accept expertise erosion and an over-reliance on AI.

Proposition 6. *For high-complexity tasks, optimal compensation α^* is monotonically decreasing in the learning rate ϕ_u and constant in the decay rate ϕ_d ; moreover, the steady-state success probability $\bar{\pi}^*$ and the firm profit $\bar{\Pi}^*$ are increasing in ϕ_u and constant in ϕ_d .*

For high-complexity tasks, in the long-run the organization settles into one of two steady states: either all workers remain non-experts, or the firm invests enough to build and maintain an all-expert workforce. In the non-expert region, learning and skill decay are irrelevant because promotions never occur. Once incentives are high enough to trigger promotions, however, the system locks into the expert state and demotions disappear. In the expert equilibrium, faster learning (ϕ_u) reduces the compensation required to sustain diligence, since non-experts can more easily become experts. This lowers the incentive cost of maintaining a high-skill workforce and therefore increases profit. Conditional on the all-expert steady state, however, performance $\bar{\pi}^* = \pi_{0,H} + K_H$ does not depend on ϕ_u . Thus faster learning does not raise success probability within that regime. Instead, ϕ_u affects performance only through regime selection: higher learning rates expand the parameter region in which the all-expert regime is optimal.

4.4 Contract Structure for Low-Complexity Tasks

Our next result shows that for low-complexity tasks, the mixed regime for worker skill always exists in steady state.

Theorem 4. *Let $\tau < 1/2$ and $\phi_u \geq \phi_d$. Then region (ii) of Theorem 2 generically exists ($h_2 < h_3$). In particular, for such values of $h \in (h_2, h_3)$, $\mu_H^* \in (0, 1)$.*

When tasks are low-complexity, skill and effort are substitutes ($\tau < 1/2$) and learning is at least as strong as skill decay ($\phi_u \geq \phi_d$). In this environment, less-skilled workers benefit more from additional diligence because there are more mistakes left to catch. As a result, incentives initially focus on non-experts rather than experts. The firm finds it cheaper to motivate lower-skill workers to exert high effort, while experts, whose baseline performance is already strong, may exert only partial effort.

This generates a richer regime structure than in the complex-task case. As stakes rise, the firm may first push non-experts to full effort while experts remain only partially diligent. Only at sufficiently high stakes does it become optimal to induce maximal effort from experts as well. Unlike the “all-or-nothing” pattern under high-complexity tasks, low-complexity tasks admit mixed long-run outcomes in which both types coexist and experts do not necessarily work at full intensity. As our baseline parameters represent this low-complexity case, these regimes were illustrated in the right-hand panel of Figure 3.

The economic intuition is that when effort substitutes for skill, organizations can compensate for limited expertise through additional diligence. This makes it possible to sustain high performance without fully committing to expert maintenance. Managerially, this implies that for low-complexity tasks, firms may rationally rely on effort from less-skilled workers and tolerate partial expert engagement, rather than investing heavily to maintain an exclusively expert workforce.

Proposition 7. *For low-complexity tasks, the expert share μ_H^* is generically non-monotone in both ϕ_d and ϕ_u . In other words, it is possible to decrease ϕ_u (resp. increase ϕ_d) and μ_H^* to increase.*

For low-complexity tasks, the equilibrium share of experts has three regimes. If the reward h is small, the firm does not pay, non-experts never go all-out, and no one is promoted, so the expert share is zero. For moderate h , the firm pays just enough so non-experts work at full tilt (they get promoted with probability ϕ_u), but not enough to keep experts at full effort, so experts sometimes slip (with probability ϕ_d); the steady state is then exactly the flow ratio $\phi_u/(\phi_u + \phi_d)$. If h is large, it becomes worthwhile to pay the minimum that keeps both states at full effort; promotions occur and demotions do not, so the expert share is one.

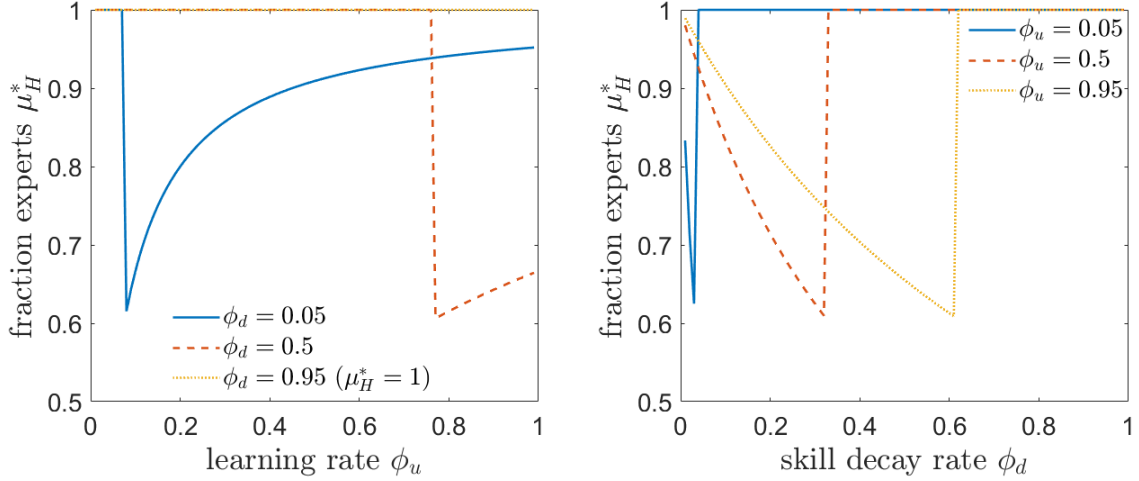


Figure 6. Fraction of staff who are experts. **Left panel:** varying rate of skill increase ϕ_u ; **Right panel:** varying rate of skill loss ϕ_d .

Therefore, increasing the learning rate ϕ_u does not generically lead to a monotone expansion of expertise. While faster learning increases upward mobility in the mixed regime, raising the flow of workers into the expert state, it simultaneously weakens expert discipline. When ϕ_u is high, slipping out of expertise is no longer catastrophic: demoted experts expect to regain their status quickly. As a result, the continuation-value loss from shirking falls, the expert full-effort incentive constraint tightens, and the bonus required to incentivize effort rises. Beyond a point, this increased cost of inducing maximal expert effort outweighs the compositional gains from faster learning, leading the firm to allow experts to ‘slip’ and operate a mixed regime instead. The left panel of Figure 6 illustrates how μ_H^* is generically non-monotonic in ϕ_u : for both $\phi_d = 0.05$ and $\phi_d = 0.5$, $\mu_H^* = 1$ when ϕ_u is low, μ_H^* drops discontinuously when the firm exits the full-effort regime and then increases gradually within the mixed regime as learning continues to improve.

Increasing the rate of skill decay ϕ_d also produces a non-monotonic organizational response (the right panel of Figure 6). When skill decay is slow, experts can reduce effort without quickly losing their skills. In this environment, firms often rely on motivated non-experts to earn promotion while tolerating partial engagement from experts. The result is a mixed workforce, with moderate quality and limited investment in sustaining top expertise. From a managerial perspective, slow skill decay can quietly justify underinvestment in expert vigilance.

As the decay rate rises to intermediate levels, however, organizations enter a fragile zone. Experts who relax effort now lose skills at a meaningful rate, yet it remains expensive to enforce constant

diligence. The steady-state share of experts falls and service quality deteriorates. This is precisely the region where firms may observe visible skill atrophy following AI adoption: employees rely on automation, expertise erodes gradually, and performance declines despite ongoing incentive pay. The problem is not that skill decay is extreme, but that it is high enough to matter and low enough that strong discipline is not yet worthwhile.

When decay becomes sufficiently rapid, the logic reverses. If shirking quickly leads to demotion and large expected income loss, expert effort becomes comparatively cheap to enforce. The firm optimally commits to strong incentives that induce full diligence, eliminate demotions, and make the expert state absorbing. Paradoxically, very high rates of skill decay can stabilize expertise because the threat of losing status becomes powerful. The organization transitions to a fully expert, high-effort steady state and restores performance.

Taken together, these results show that both learning and skill decay can generate non-monotonic organizational adjustments. Faster learning does not always strengthen expertise, and faster decay does not always undermine it. What matters is how these forces alter workers' dynamic incentives and the cost of sustaining effort. Rather than gradual changes in workforce composition, firms may experience regime shifts, for example, moving from a mixed, eroding organization to a fully expert one once incentive thresholds are crossed. For managers, this implies that modest increases in skill decay can be far more destabilizing than extreme ones, and that sustaining expertise in AI-enabled environments requires anticipating these tipping points rather than reacting to slow declines.

Proposition 8. *Generically, for low-complexity tasks, the optimal compensation α^* is non-monotonic in both ϕ_u and ϕ_d . Moreover, profit $\bar{\Pi}^*$ and average output success $\bar{\pi}^*$ are also generically non-monotonic in ϕ_u and ϕ_d .*

As ϕ_u rises, learning becomes easier, strengthening the extensive margin by making promotions from non-experts more likely when effort is high. When incentives are interior, this raises the return to incentivizing non-experts, leading the firm to increase α . However, when effort and skill are substitutes, faster learning also weakens discipline among experts: if they shirk, they expect to regain expertise quickly. This tightens the expert incentive constraint and raises the cost of sustaining full expert effort. Beyond this point, the firm abandons maximal expert incentives and switches to a regime in which non-experts are incentivized while experts exert only partial effort. As a result, output and profit need not rise monotonically with ϕ_u . This contrasts with Proposition 6, where effort and skill are complements and stronger learning uniformly lowers incentive costs without undermining discipline.

As the decay rate ϕ_d increases, its impact is also non-monotonic. At low to moderate levels, faster skill decay makes promotions less durable and weakens the long-run expert base. Firms must pay higher bonuses to sustain effort, reducing output and profit. In this region, modest increases in skill decay can quietly erode organizational capability and raise oversight costs.

When decay becomes sufficiently rapid, however, the logic reverses. If shirking quickly leads to demotion and income loss, expert effort becomes easier to enforce. Both types work at full effort, the required bonus falls, and performance stabilizes or even improves. Paradoxically, severe skill decay can discipline experts and reduce agency losses once strong incentives are in place.

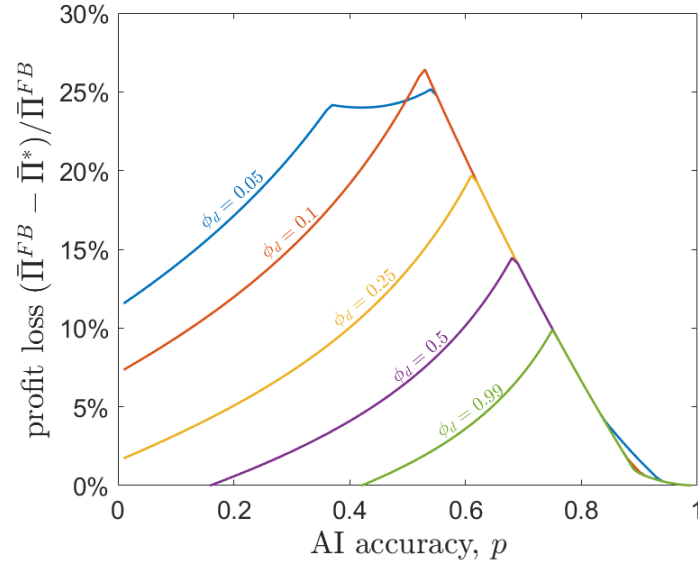


Figure 7. Percentage loss of profit due to agency problem, varying p and ϕ_d , and other parameters at baseline

Figure 7 illustrates this pattern, showing that when ϕ_d is low, increasing ϕ_d can reduce profits relative to first best, e.g., moving from $\phi_d = 0.05$ to $\phi_d = 0.1$ when $p = 0.5$ increases the profit loss. The overall trend in Figure 7, however, is that increasing ϕ_d increases profits relative to first best. In addition, the largest performance losses occur at intermediate AI accuracy, where oversight and skill maintenance are most costly. Managerially, the key lesson is that moderate skill decay is more destabilizing than extreme rates of decay: small increases in skill decay can undermine capability unless firms fully commit to sustaining effort.

5 Conclusion

In an address to the priests of the Diocese of Rome, Pope Leo XIV warned of the temptation to use AI when writing homilies: “Like all the muscles in the body, if we do not use them, if we do not move them, they die. The brain needs to be used, so our intelligence must also be exercised a little so as not to lose this capacity” (Cernuzio, 2026). In this paper, we study how this risk of AI-induced atrophy influences incentives. Our central insight is that AI adoption is not merely a technological upgrade, but an organizational design problem: improvements in machine accuracy reshape the marginal value of human diligence, while incentive choices today determine whether expertise is preserved or quietly erodes tomorrow. When effort corrects current errors *and* sustains future skill, standard intuition about incentives, pay, and technology adoption can fail.

In our principal-agent model, the firm can incentivize output but cannot directly enforce worker effort to engage with AI. We find that under these conditions, intermediate AI reliability is a ‘danger zone’: systems are good enough that errors are infrequent, but not good enough to be ignored, making shirking tempting and incentivizing vigilance expensive. Therefore, strong incentives are needed to ensure short-term quality. In general, incremental improvements in AI can raise oversight costs and

worsen organizational performance if incentive design does not adjust accordingly.

When active engagement with AI builds and maintains expertise, then absent contracting frictions the firm prefers that workers remain engaged and maintain skills. When the firm relies on output incentives, however, it is often optimal to let skills degrade. The impact of the rates of skill learning and loss depend on the complexity of the task. When tasks are complex, with long training periods and high rates of skill decay, then as expected, faster learning increases profits. When tasks are lower-complexity, however, the negative impact of the contracting friction can decline when the risk of skill loss increases or when it requires more effort to regain lost skills. In such an environment, workers are self-motivated to maintain their skills, and it is less expensive for the firm to ensure expertise.

The value, or impact, of the output determines which mix of skills is optimal for the firm; an AI-enabled customer service agent in a retailer's call center will operate in a different regime than an emergency physician consulting an AI. If the output value is relatively low as compared to the cost of labor, as in some call centers, the firm can let skills degrade and workers pass along AI output with little oversight. At moderate values, for low-complexity tasks, the firm may prefer a mix of non-experts who are highly engaged and experts who put in less effort. At the highest levels of value, as for that emergency physician, the firm should provide incentives to ensure that the human is always engaged and highly skilled.

Our findings offer insights for managers adopting AI. First, organizations should anticipate that intermediate AI reliability requires the highest oversight investment, even when the technology is improving. Second, incentive structures should reflect whether tasks require expert judgment (high-complexity tasks) or can be done through diligence (low-complexity tasks): the same AI system may require different contracts across roles. Third, firms should explicitly account for expertise erosion when evaluating the ROI of AI tools; failure to do so can create hidden vulnerabilities or capability traps. In many settings, it may be optimal to directly monitor skill and effort, through testing, rotation, or managerial attention, precisely when AI performance appears strong. In general, effective AI deployment requires not only technological adoption but also careful redesign of incentive systems to sustain human expertise.

Several extensions naturally fit within our modeling framework. Allowing for teams or layered oversight would clarify how the effects of task complexity propagates across organizational hierarchies, potentially amplifying the all-or-nothing logic we identify. Endogenizing AI investment would formalize the trade-off between spending on training better models versus stronger incentives, which our results already suggest depends critically on the effort-skill interaction. Incorporating behavioral frictions or career concerns would modify incentive constraints, but would operate through the same mechanism: changing how current diligence translates into future expertise. Embedding our model within a competitive framework would allow us to examine industry dynamics, as in [Besanko et al. \(2010\)](#). In our model we assume that the principal offers a single time-invariant, piece-rate incentive for the agents; these are robust, renegotiation-proof contracts ([Gibbons and Roberts, 2013](#)). That said, we may also consider contracts that incorporate noisy signals of worker output, as well as more complex dynamic contracts as in ([Sun and Tian, 2018](#)). Finally, empirical work in domains such as medical review, auditing, or customer support could use our framework to identify learning and skill

decay dynamics and test the model's non-monotonic incentive predictions.

More broadly, this paper offers a theory of why organizations can appear to function well with AI while becoming increasingly brittle underneath. AI changes what workers can do; incentives determine what they choose to do; and learning and expertise decay determine what they will be able to do tomorrow. Leaders who ignore any one of these effects risks mistaking short-run gains for sustainable performance.

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A Proofs

We present all omitted proofs from the main body.

A.1 Proofs from Section 3

Proof of Proposition 1. The organization chooses effort $x \in [0, 1]$ to solve $\max_{x \in [0, 1]} h\pi(s, x) - \frac{1}{2}cx^2$. Collecting terms shows that $\pi(s, x)$ is affine in x , $\pi(s, x) = a(s) + K(s)x$, where

$$a(s) = p + (1 - p)\theta s(1 - \tau), \quad K(s) = (1 - p)q[(1 - \tau)(1 - \theta s) + \tau\theta s].$$

Since all primitives lie in $[0, 1]$, $K(s) \geq 0$. Substituting yields the strictly concave objective

$$ha(s) + hK(s)x - \frac{1}{2}cx^2,$$

whose unconstrained maximizer satisfies $hK(s) - cx = 0$, giving $\tilde{x}^{FB}(s) = hK(s)/c$. Imposing $x \in [0, 1]$ gives

$$x^{FB}(s) = (hK(s)/c)_0^1 \equiv \min\{\max\{hK(s)/c, 0\}, 1\}.$$

Substituting into π gives $\pi^{FB}(s) = a(s) + K(s)x^{FB}(s)$ and

$$\Pi^{FB}(s) = h\pi^{FB}(s) - \frac{1}{2}c(x^{FB}(s))^2.$$

To establish comparative statics, define $f(x; p, s, q) = h\pi(s, x) - \frac{1}{2}cx^2$. Using $\pi(s, x) = 1 - (1 - p)B(s, x)$ with $B(s, x) \geq 0$, we have $\partial\pi/\partial p = B(s, x) \geq 0$. Direct differentiation yields

$$\frac{\partial\pi}{\partial s} = (1 - p)\theta[(1 - \tau)(1 - qx) + \tau qx] \geq 0, \quad \frac{\partial\pi}{\partial q} = (1 - p)x[(1 - \tau)(1 - \theta s) + \tau\theta s] \geq 0.$$

Hence for any fixed x , $f(x; p, s, q)$ is weakly increasing in (p, s, q) . Let $F(p, s, q) = \max_{x \in [0, 1]} f(x; p, s, q)$. Since the pointwise maximum of functions that are increasing in a parameter is also increasing, F is weakly increasing in each of p, s, q . Noting that $F(p, s, q) = \Pi^{FB}$ establishes the result. $\square$

Proof of Theorem 1. Fix skill $s \in [0, 1]$. From (1), success probability is affine in effort, $\pi(s, x) = \pi_0 + Kx$, where

$$\pi_0 \equiv 1 - (1 - p)[(1 - \tau)(1 - \theta s) + \tau], \quad K \equiv (1 - p)q[(1 - \tau)(1 - \theta s) + \tau\theta s] \geq 0.$$

The worker chooses $x \in [0, 1]$ to maximize

$$u(\alpha; x) = \alpha(\pi_0 + Kx) - \frac{1}{2}cx^2.$$

The first-order condition yields $x = \alpha K/c$, so the best response is

$$x^*(\alpha) = (\alpha K/c)_0^1.$$

Firm profit is

$$\Pi(\alpha) = (h - \alpha)(\pi_0 + Kx^*(\alpha)), \quad \alpha \geq 0.$$

We analyze three regions:

(i) *Interior region.* If $0 < x^* < 1$ (equivalently $0 < \alpha < c/K$), then $x^*(\alpha) = \alpha K/c$ and

$$\Pi(\alpha) = (h - \alpha) \left(\pi_0 + \frac{K^2}{c} \alpha \right).$$

Let $B \equiv K^2/c$. Then $\Pi'(\alpha) = -\pi_0 + Bh - 2B\alpha$, giving the unique maximizer

$$\alpha^\circ = \frac{Bh - \pi_0}{2B} = \frac{h}{2} - \frac{c\pi_0}{2K^2}.$$

This solution is interior iff

$$0 < \alpha^\circ < c/K \iff \underline{h} < h < \bar{h},$$

where $\underline{h} \equiv c\pi_0/K^2$ and $\bar{h} \equiv \underline{h} + 2c/K$. The corresponding effort, success probability, and profit are

$$x^* = \frac{\alpha^\circ K}{c} = \frac{1}{2} \left(\frac{hK}{c} - \frac{\pi_0}{K} \right), \quad \pi^* = \frac{\pi_0}{2} + \frac{hK^2}{2c},$$

$$\Pi^* = \frac{h\pi_0}{2} + \frac{h^2 K^2}{4c} + \frac{c\pi_0^2}{4K^2}.$$

(ii) *Lower corner.* If $h \leq \underline{h}$, the interior solution satisfies $\alpha^\circ \leq 0$, so the firm sets $\alpha^* = 0$. Then $x^* = 0$, $\pi^* = \pi_0$, and $\Pi^* = h\pi_0$.

(iii) *Upper corner.* If $h \geq \bar{h}$, the interior solution satisfies $\alpha^\circ \geq c/K$. To induce $x^* = 1$ the firm must set $\alpha \geq c/K$, and since $\Pi(\alpha) = (h - \alpha)(\pi_0 + K)$ is decreasing in α , the optimal choice is $\alpha^* = c/K$. Thus $x^* = 1$, $\pi^* = \pi_0 + K$, and $\Pi^* = (h - c/K)(\pi_0 + K)$.

Combining the three regions yields the optimal contract, effort, success probability, and profit stated in Theorem 1. $\square$

Proof of Proposition 2. From Theorem 1, the optimal piece rate is

$$\alpha^* = \begin{cases} 0, & h \leq \underline{h}, \\ \frac{h}{2} - \frac{c\pi_0}{2K^2}, & \underline{h} < h < \bar{h}, \\ \frac{c}{K}, & h \geq \bar{h}, \end{cases} \quad \underline{h} = \frac{c\pi_0}{K^2}, \quad \bar{h} = \underline{h} + \frac{2c}{K}.$$

We repeatedly use

$$\partial_p \pi_0 > 0, \quad \partial_p K < 0, \quad \partial_q \pi_0 = 0, \quad \partial_q K > 0,$$

and

$$\partial_s \pi_0 = (1 - p)\theta(1 - \tau) \geq 0, \quad \partial_s K = (1 - p)q\theta(2\tau - 1).$$

(a) Variation in p . Since π_0 increases and K decreases in p , both thresholds $\underline{h} = c\pi_0/K^2$ and $\bar{h} = \underline{h} + 2c/K$ increase in p . Hence, as p rises (holding h fixed), the equilibrium region can only move from upper corner $\rightarrow$ interior $\rightarrow$ no-pay.

Within regions,

$$\alpha^*(p) = \begin{cases} c/K, & \text{upper corner,} \\ \frac{h}{2} - \frac{c}{2} \frac{\pi_0}{K^2}, & \text{interior,} \\ 0, & \text{no-pay.} \end{cases}$$

Thus α^* increases in p in the upper corner, decreases in p in the interior, and is constant in the no-pay region. Because region transitions occur only when $h = \bar{h}(p)$ or $h = \underline{h}(p)$, $\alpha^*(p)$ is single-peaked. Let $p^*(\tau)$ solve $h = \bar{h}(p, \tau)$. Since $\partial_p \bar{h} > 0$,

$$\frac{dp^*}{d\tau} = -\frac{\partial_\tau \bar{h}}{\partial_p \bar{h}}.$$

Differentiating

$$\bar{h} = \frac{c\pi_0}{K^2} + \frac{2c}{K}$$

gives

$$\partial_\tau \bar{h} = \frac{c}{K^2} \partial_\tau \pi_0 - \left(\frac{2c\pi_0}{K^3} + \frac{2c}{K^2} \right) \partial_\tau K.$$

Using $\partial_\tau \pi_0 = -(1-p)\theta s$ and $\partial_\tau K = (1-p)q(2\theta s - 1)$ yields

$$\text{sign}(\partial_\tau \bar{h}) = \text{sign}(-K\theta s - (2\pi_0 + 2K)q(2\theta s - 1)).$$

Hence $dp^*/d\tau > 0$ if $\theta s > 1/2$, while the opposite holds for sufficiently small s .

(b) Variation in q . Because π_0 does not depend on q and K increases in q , both thresholds decrease in q . Thus the economy moves no-pay $\rightarrow$ interior $\rightarrow$ upper corner as q rises.

Within regions,

$$\alpha^*(q) = 0, \quad \alpha^*(q) = \frac{h}{2} - \frac{c}{2} \frac{\pi_0}{K^2}, \quad \alpha^*(q) = \frac{c}{K},$$

respectively, implying α^* rises in the interior and falls in the upper corner. Hence $\alpha^*(q)$ is single-peaked at $h = \bar{h}(q)$.

Let $q^*(\tau)$ solve $h = \bar{h}(q, \tau)$. Since $\partial_q \bar{h} < 0$,

$$\text{sign}\left(\frac{dq^*}{d\tau}\right) = -\text{sign}\left(\frac{\partial_\tau \bar{h}}{\partial_q \bar{h}}\right) = \text{sign}(\partial_\tau \bar{h}),$$

which is the opposite pattern relative to $dp^*/d\tau$.

(c) Variation in s . In the no-pay region $\alpha^*(s) = 0$. In the upper corner,

$$\alpha^*(s) = \frac{c}{K}, \quad \partial_s \alpha^* = -\frac{c}{K^2} \partial_s K,$$

which is negative under complements ($\tau > 1/2$) and positive under substitutes ($\tau < 1/2$).

In the interior,

$$\alpha^*(s) = \frac{h}{2} - \frac{c}{2} \frac{\pi_0(s)}{K(s)^2},$$

so

$$\partial_s \alpha^* = -\frac{c}{2K(s)^3} \left(K(s) \partial_s \pi_0(s) - 2\pi_0(s) \partial_s K(s) \right).$$

Here

$$\partial_s \pi_0(s) = (1-p)(1-\tau)\theta > 0, \quad \partial_s K(s) = (1-p)q\theta(2\tau-1).$$

Thus, when $\tau > 1/2$, the second term works to increase α^* while the first works in the opposite direction. Accordingly, the sign of $\partial_s \alpha^*$ in the interior is not determined by $\tau > 1/2$ alone; it is positive on an open set of primitives for which

$$2\pi_0(s) \partial_s K(s) > K(s) \partial_s \pi_0(s).$$

Likewise, when $\tau < 1/2$, the reverse force operates and $\partial_s \alpha^*$ is negative on an open set of primitives. Combined with the opposite slope in the upper-corner region, this yields the generic single-peakedness under complements and generic single-troughedness under substitutes stated in the proposition. $\square$

Proof of Proposition 3. Recall the optimal profit levels from Theorem 1:

$$\Pi^* = \begin{cases} h\pi_0, & h \leq \underline{h}, \\ \frac{h\pi_0}{2} + \frac{h^2 K^2}{4c} + \frac{c\pi_0^2}{4K^2}, & \underline{h} < h < \bar{h}, \\ (h - \frac{c}{K})(\pi_0 + K), & h \geq \bar{h}. \end{cases}$$

(a) Variation in q . Since π_0 does not depend on q while $\partial_q K > 0$, it suffices to check $\partial \Pi^* / \partial K$. In the interior,

$$\frac{\partial \Pi^*}{\partial K} = \frac{h^2}{2c} K - \frac{c\pi_0^2}{2K^3}.$$

Interior feasibility implies $h > \underline{h} = c\pi_0/K^2$, which yields $\partial \Pi^* / \partial K > 0$. In the upper corner,

$$\frac{\partial \Pi^*}{\partial K} = \frac{c}{K^2}(\pi_0 + K) + \left(h - \frac{c}{K} \right) > 0.$$

In the lower corner $\Pi^* = h\pi_0$ is independent of q . Because both thresholds $\underline{h}, \bar{h}$ fall with q , increases in q can only move the economy toward regions where profit strictly rises with q . Hence Π^* is globally increasing in q .

(b) Variation in p . Here $\partial_p \pi_0 > 0$ and $\partial_p K < 0$. In the interior,

$$\frac{d\Pi^*}{dp} = \left(\frac{h}{2} + \frac{c\pi_0}{2K^2} \right) \partial_p \pi_0 + \left(\frac{h^2}{2c} K - \frac{c\pi_0^2}{2K^3} \right) \partial_p K.$$

The first term is positive while the second is negative, reflecting a positive “baseline” effect and a

negative “effort-leverage” effect. For sufficiently small c , the latter dominates over an interior range, so $d\Pi^*/dp < 0$. However, in the lower corner $\Pi^* = h\pi_0$ so $d\Pi^*/dp > 0$, while in the upper corner the derivative can again be positive for large h . Thus $\Pi^*(p)$ is generically non-monotone.

(c) Variation in s . Note $\partial_s \pi_0 \geq 0$ and $\partial_s K = (1-p)q\theta(2\tau-1)$.

If $\tau > 1/2$ (complements), then $\partial_s K > 0$. Since $\partial\Pi^*/\partial\pi_0 > 0$ and $\partial\Pi^*/\partial K > 0$ in both the interior and upper corner,

$$\frac{d\Pi^*}{ds} = \frac{\partial\Pi^*}{\partial\pi_0}\partial_s\pi_0 + \frac{\partial\Pi^*}{\partial K}\partial_s K > 0.$$

Hence $\Pi^*(s)$ is weakly increasing.

If $\tau < 1/2$ (substitutes), then $\partial_s K < 0$. In the interior,

$$\frac{d\Pi^*}{ds} = A(s) - C(s),$$

where

$$A(s) = \left(\frac{h}{2} + \frac{c\pi_0}{2K^2}\right)\partial_s\pi_0 \geq 0, \quad C(s) = \left(\frac{h^2}{2c}K - \frac{c\pi_0^2}{2K^3}\right)|\partial_s K| \geq 0.$$

When effort is cheap (small c), the coefficient in $C(s)$ becomes large, so $C(s) > A(s)$ over an interior range and $d\Pi^*/ds < 0$ there. Since in the lower corner $\Pi^* = h\pi_0$ implies $d\Pi^*/ds \geq 0$, profit initially rises with s but falls over a middle interval. Thus $\Pi^*(s)$ is generically non-monotone under substitutes.

□

A.2 Proofs from Section 4

Proof of Proposition 4. Fix a stationary piece rate $\alpha \geq 0$. The worker’s period payoff is bounded because $x \in [0, 1]$ and $\pi(s, x) \in [0, 1]$, so $|u(s, x; \alpha)| \leq \alpha + \frac{c}{2}$. Hence, for any admissible effort path, the discounted payoff $\sum_{t \geq 1} \delta^{t-1} u(s_t, x_t; \alpha)$ is absolutely convergent, and $\bar{U}(\alpha)$ is well defined.

Let $V^\alpha = (V_L^\alpha, V_H^\alpha)$ denote the worker’s value function. Under the transition rule in Section 4, the Bellman operator $T_\alpha : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is

$$(T_\alpha V)_L = \max_{x \in [0, 1]} \left\{ u(L, x; \alpha) + \delta[(1 - \phi_u \mathbf{1}_{\{x=1\}})V_L + \phi_u \mathbf{1}_{\{x=1\}}V_H] \right\},$$

$$(T_\alpha V)_H = \max_{x \in [0, 1]} \left\{ u(H, x; \alpha) + \delta[\phi_d \mathbf{1}_{\{x < 1\}}V_L + (1 - \phi_d \mathbf{1}_{\{x < 1\}})V_H] \right\}.$$

Because rewards are bounded and the continuation term is discounted by $\delta < 1$,

$$\|T_\alpha V - T_\alpha W\|_\infty \leq \delta \|V - W\|_\infty,$$

so T_α is a contraction. By the Banach fixed-point theorem, for each α there exists a unique fixed point V^α .

We next characterize optimal actions. Since $\pi(s, x) = \pi_{0,s} + K_s x$, the stage payoff is

$$u(s, x; \alpha) = \alpha\pi_{0,s} + \alpha K_s x - \frac{c}{2}x^2.$$

For a given continuation vector V , the dependence of the Bellman objective on x is quadratic on the region $x < 1$, while the continuation payoff changes only at the threshold $x = 1$. Therefore, in each state the maximizer must be obtained by comparing at most two candidates: full effort $x = 1$, and the maximizer of the quadratic part on $[0, 1)$.

For state L , if $x < 1$ then no promotion occurs, so the continuation value is δV_L and the worker solves

$$\max_{x \in [0,1)} \alpha \pi_{0,L} + \alpha K_L x - \frac{c}{2} x^2 + \delta V_L.$$

Thus the unique optimizer on $[0, 1)$ is $x_L^\circ(\alpha) = \alpha K_L / c$ when $\alpha K_L / c < 1$; if $\alpha K_L / c \geq 1$, the objective is increasing on $[0, 1)$ and its supremum is attained at the boundary $x \uparrow 1$. Hence an optimal action in state L is obtained by comparing

$$x = x_L^\circ(\alpha) \quad (\text{when } \alpha K_L / c < 1) \quad \text{and} \quad x = 1.$$

Analogously, in state H , if $x < 1$ the continuation value is $\delta[\phi_d V_L + (1 - \phi_d) V_H]$, so the unique optimizer on $[0, 1)$ is $x_H^\circ(\alpha) = \alpha K_H / c$ when $\alpha K_H / c < 1$, and otherwise the supremum over $[0, 1)$ is approached as $x \uparrow 1$. Hence an optimal action in state H is obtained by comparing

$$x = x_H^\circ(\alpha) \quad (\text{when } \alpha K_H / c < 1) \quad \text{and} \quad x = 1.$$

Therefore, for each α , an optimal stationary Markov action exists. Moreover, except when one of these pairwise comparisons holds with equality, the optimal action in each state is unique. These indifference conditions are scalar equalities in the primitives, and therefore arise only on a knife-edge (measure-zero) subset of the parameter space. Thus the worker's stationary Markov best response is unique except on a set of measure zero.

Given any stationary Markov policy $x^\alpha(\cdot)$, the induced skill process is a time-homogeneous Markov chain on the finite state space $\{L, H\}$. Starting from $s_1 = L$, this chain always enters a unique recurrent class: depending on whether $x^\alpha(L)$ and $x^\alpha(H)$ are equal to 1 or strictly below 1, the recurrent class is either $\{L\}$, $\{H\}$, or the irreducible two-state class generated by promotion and demotion. Since per-period firm payoff

$$r^\alpha(s) = (h - \alpha)\pi(s, x^\alpha(s))$$

is bounded, standard finite-state Markov reward process results imply that

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T r^\alpha(s_t)$$

exists almost surely and equals the average reward of that recurrent class. Hence $\bar{\Pi}(\alpha)$ is well defined.

Finally, the firm chooses α . Because paying $\alpha > h$ yields weakly negative payoff per successful outcome, while $\alpha = h$ yields zero per successful outcome, no optimizer can satisfy $\alpha > h$; thus it suffices to restrict attention to the compact interval $[0, h]$. For each fixed stationary worker regime, the transition matrix is constant and $r^\alpha(s)$ is affine or quadratic in α , so the induced steady-state profit

is continuous and piecewise smooth in α . Since there are only finitely many worker regimes, $\bar{\Pi}(\alpha)$ is continuous on $[0, h]$, and an optimal α exists by Weierstrass. On each regime the objective has at most one interior maximizer, so multiplicity can arise only when two regime-specific branches yield exactly the same profit, again a knife-edge condition. Hence the Markov-perfect equilibrium exists, and is unique except for a set of parameters of measure zero. $\square$

Proof of Proposition 5. The proof proceeds in three steps.

1. *Step 1* defines the three natural steady-state policies (LL, LH, HH) and their profits.
2. *Step 2* shows that the dynamic first-best reduces to comparing these policies and characterizes the regime cutoffs.
3. *Step 3* proves monotonicity of steady-state first-best profit in (ϕ_u, ϕ_d) .

Step 1: Candidate steady-state policies. Under stationary policies the induced skill process is a finite-state Markov chain. Because per-period payoffs depend only on current skill and effort, each stationary policy yields a constant steady-state profit. It therefore suffices to compare the following policies.

(i) **Policy LL.** Low-skill workers exert interior first-best effort $x_L^{FB}(h) = hK_L/c < 1$, so promotion never occurs. The steady state is L and per-period profit equals

$$\Pi_{LL}^{FB}(h) = h\pi_{0,L} + \frac{h^2 K_L^2}{2c}. \quad (3)$$

(ii) **Policy LH.** Low types exert full effort $x_L = 1$ and are promoted with probability ϕ_u , while high types exert interior effort $x_H^{FB}(h) = hK_H/c < 1$ and are demoted with probability ϕ_d . The stationary distribution is

$$\mu_L = \frac{\phi_d}{\phi_u + \phi_d}, \quad \mu_H = \frac{\phi_u}{\phi_u + \phi_d}. \quad (4)$$

Per-period profits conditional on skill are

$$a_L(h) = h(\pi_{0,L} + K_L) - \frac{c}{2}, \quad a_H(h) = h\pi_{0,H} + \frac{h^2 K_H^2}{2c}. \quad (5)$$

Thus steady-state profit equals

$$\Pi_{LH}^{FB}(h) = \mu_L a_L(h) + \mu_H a_H(h). \quad (6)$$

(iii) **Policy HH.** Both types exert full effort $x_L = x_H = 1$, the chain absorbs at H , and per-period profit equals

$$\Pi_{HH}^{FB}(h) = h(\pi_{0,H} + K_H) - \frac{c}{2}. \quad (7)$$

Step 2: Profit comparisons and regime cutoffs. The dynamic first-best satisfies

$$\Pi^{FB}(h) = \max \{ \Pi_{LL}^{FB}(h), \Pi_{LH}^{FB}(h), \Pi_{HH}^{FB}(h) \}. \quad (8)$$

We begin with the comparison between LL and LH. Define

$$\Delta_1(h) = \Pi_{LH}^{FB}(h) - \Pi_{LL}^{FB}(h). \quad (9)$$

Using (3)–(6), $\Delta_1(h)$ is a quadratic function of h . Since $\Pi_{LH}^{FB}(0) = -\mu_L c/2 < 0$ while $\Pi_{LL}^{FB}(0) = 0$, we have $\Delta_1(0) < 0$. Hence LL is optimal for sufficiently small h .

Let $h_H \equiv c/K_H$ denote the static cutoff at which the expert's first-best effort switches from interior to full effort.

Case A. Suppose $\Delta_1(h) \leq 0$ for all $h < h_H$. Then the mixed policy LH never dominates LL on the region where expert effort is interior. For $h \geq h_H$, expert effort is optimally full, so HH weakly dominates LH. Hence the planner compares only LL and HH. Define

$$\Delta_{HL}(h) = \Pi_{HH}^{FB}(h) - \Pi_{LL}^{FB}(h). \quad (10)$$

Using (3) and (7), $\Delta_{HL}(h)$ is a quadratic in h with negative leading coefficient and $\Delta_{HL}(0) < 0$. The LL expression is economically relevant only while the non-expert effort remains interior, i.e., $x_L^{FB}(h) = hK_L/c < 1$, which restricts the admissible domain to $h < c/K_L$. Over this domain the concavity of $\Delta_{HL}(h)$ implies that it can cross zero at most once. Since $\Delta_{HL}(0) < 0$ and $\Delta_{HL}(c/K_L) > 0$, there exists a unique economically relevant positive root, denoted h^{FB*} . Therefore

$$\begin{cases} h < h^{FB*} & \Rightarrow \Pi^{FB}(h) = \Pi_{LL}^{FB}(h), \\ h > h^{FB*} & \Rightarrow \Pi^{FB}(h) = \Pi_{HH}^{FB}(h). \end{cases}$$

Case B. Suppose instead that $\Delta_1(h)$ becomes positive for some $h < h_H$. Let h^{FB} denote the smallest positive root of $\Delta_1(h) = 0$. Then

$$\Pi_{LL}^{FB}(h) > \Pi_{LH}^{FB}(h) \quad \text{for } h < h^{FB},$$

while

$$\Pi_{LH}^{FB}(h) > \Pi_{LL}^{FB}(h) \quad \text{for } h \in (h^{FB}, h_H).$$

Next compare HH and LH. Define

$$\Delta_2(h) = \Pi_{HH}^{FB}(h) - \Pi_{LH}^{FB}(h). \quad (11)$$

From (6)–(7), $\Delta_2(0) < 0$. The interior LH expression is only relevant while the expert effort remains interior, i.e., $x_H^{FB}(h) = hK_H/c < 1$. Thus the LH branch is defined only for $h < c/K_H$. Evaluating the profit difference at the boundary $h = c/K_H$ yields $\Delta_2(c/K_H) > 0$. Since $\Delta_2(h)$ is continuous on $[0, c/K_H]$, there exists a smallest positive root, denoted $\bar{h}^{FB}$.

If $\bar{h}^{FB} \leq h^{FB}$ the LH region disappears, which collapses to Case A. Otherwise $h^{FB} < \bar{h}^{FB}$ and the

planner's policy has three regimes:

$$\begin{cases} 0 < h < h^{FB} & \Rightarrow \Pi^{FB}(h) = \Pi_{LL}^{FB}(h), \\ h^{FB} < h < \bar{h}^{FB} & \Rightarrow \Pi^{FB}(h) = \Pi_{LH}^{FB}(h), \\ h > \bar{h}^{FB} & \Rightarrow \Pi^{FB}(h) = \Pi_{HH}^{FB}(h). \end{cases}$$

Step 3: Monotonicity in (ϕ_u, ϕ_d) . Only the mixed policy depends on (ϕ_u, ϕ_d) . Using (6) we write

$$\Pi_{LH}^{FB}(h) = a_L(h) + \mu_H(a_H(h) - a_L(h)), \quad (12)$$

where $\mu_H = \phi_u/(\phi_u + \phi_d)$. We first show

$$a_H(h) \geq a_L(h) \quad \text{for all } h > 0. \quad (13)$$

For any $x \in [0, 1]$, we know $\pi(H, x) \geq \pi(L, x)$. Hence

$$h\pi(H, x) - \frac{c}{2}x^2 \geq h\pi(L, x) - \frac{c}{2}x^2.$$

Because $a_L(h) = u(L, 1)$ and $a_H(h) = \max_x u(H, x)$, inequality (13) follows.

Using (12),

$$\frac{\partial \Pi_{LH}^{FB}}{\partial \mu_H} = a_H(h) - a_L(h) \geq 0.$$

Since μ_H increases in ϕ_u and decreases in ϕ_d ,

$$\frac{\partial \Pi_{LH}^{FB}}{\partial \phi_u} \geq 0, \quad \frac{\partial \Pi_{LH}^{FB}}{\partial \phi_d} \leq 0.$$

The profits (3) and (7) do not depend on (ϕ_u, ϕ_d) . Because the maximum in (8) preserves these monotonicities, $\Pi^{FB}(h)$ is weakly increasing in ϕ_u and weakly decreasing in ϕ_d . $\square$

Proof of Theorem 2. The proof proceeds in three steps. First, we define the two worker-side bonus thresholds $0 < \underline{\alpha} < \bar{\alpha}$ that partition the firm's problem into three implementable bonus regions. Second, on each region we characterize the induced steady-state skill distribution, success probability, and firm profit, and solve the firm's within-region problem. Third, we compare the resulting region-specific profit functions to define the cutoffs $0 < h_1 \leq h_2 \leq h_3$, and then verify the final claim that $h_3 < \bar{h}_L$, where $\bar{h}_L$ is the static full-effort cutoff for a fixed L -type worker.

Step 1: Worker-side thresholds and implementable regions. Fix a stationary bonus $\alpha \geq 0$. Let $x_s^*(\alpha)$ denote the worker's equilibrium effort in state $s \in \{L, H\}$. Define

$$\underline{\alpha} := \inf\{\alpha \geq 0 : x_L^*(\alpha) = 1\},$$

the smallest bonus that pins the low-skill worker at full effort, and define

$$\bar{\alpha} := \inf\{\alpha \geq \underline{\alpha} : x_H^*(\alpha) = 1\},$$

the smallest bonus that, conditional on L already being pinned, also pins the high-skill worker. These thresholds partition the firm's problem into three steady-state relevant regions:

- R1.** $\alpha < \underline{\alpha}$: low-skill workers do not exert full effort, so starting from $s_1 = L$ promotions never occur and the steady state is all non-expert;
- R2.** $\underline{\alpha} \leq \alpha < \bar{\alpha}$: low-skill workers exert full effort while high-skill workers exert interior effort, so the steady state is mixed;
- R3.** $\alpha \geq \bar{\alpha}$: both skill types exert full effort, so the steady state is all expert.

Thus the firm needs only compare the three corresponding steady-state profit functions.

Step 2: Firm profit by bonus region. Region R1: all non-expert, interior effort. If $\alpha < \underline{\alpha}$, then

$$x_L^*(\alpha) = \frac{\alpha K_L}{c} \in [0, 1), \quad x_H^*(\alpha) = \frac{\alpha K_H}{c} \in [0, 1).$$

Starting from $s_1 = L$, promotions never occur, so the skill process remains at L forever. Hence steady-state success probability is

$$\bar{\pi}_{LL}(\alpha) = \pi_{0,L} + \frac{K_L^2}{c}\alpha, \quad (14)$$

and the firm's steady-state profit is

$$\bar{\Pi}_{LL}(\alpha; h) = (h - \alpha) \left(\pi_{0,L} + \frac{K_L^2}{c}\alpha \right). \quad (15)$$

The objective in (15) is a strictly concave quadratic in α , with unconstrained maximizer

$$\alpha_{LL}^\circ(h) = \frac{h}{2} - \frac{c\pi_{0,L}}{2K_L^2}. \quad (16)$$

Evaluating the derivative of (15) at $\alpha = 0$ gives

$$\frac{\partial \bar{\Pi}_{LL}}{\partial \alpha}(0; h) = h \frac{K_L^2}{c} - \pi_{0,L},$$

so the lower cutoff is

$$h_1 = \frac{c\pi_{0,L}}{K_L^2}. \quad (17)$$

Hence if $h < h_1$, the firm chooses $\alpha^* = 0$, while if $h > h_1$ the best bonus in Region R1 is (16), provided it remains below $\underline{\alpha}$. Substituting (16) into (14)–(15) yields

$$\bar{\pi}^* = \frac{\pi_{0,L}}{2} + \frac{hK_L^2}{2c}, \quad \bar{\Pi}^* = \left(\frac{h}{2} + \frac{c\pi_{0,L}}{2K_L^2} \right) \left(\frac{\pi_{0,L}}{2} + \frac{hK_L^2}{2c} \right) \quad (18)$$

whenever Region R1 is optimal with positive interior effort.

Region R2: mixed regime. If $\underline{\alpha} \leq \alpha < \bar{\alpha}$, then

$$x_L^*(\alpha) = 1, \quad x_H^*(\alpha) = \frac{\alpha K_H}{c} \in (0, 1).$$

The induced Markov chain over $\{L, H\}$ is irreducible, with stationary distribution

$$\mu_L = \frac{\phi_d}{\phi_u + \phi_d}, \quad \mu_H = \frac{\phi_u}{\phi_u + \phi_d}. \quad (19)$$

Therefore steady-state success probability is affine in α :

$$\bar{\pi}_{LH}(\alpha) = \mu_L(\pi_{0,L} + K_L) + \mu_H \left(\pi_{0,H} + \frac{K_H^2}{c} \alpha \right) = A + B\alpha, \quad (20)$$

where

$$A = \frac{\phi_d}{\phi_u + \phi_d}(\pi_{0,L} + K_L) + \frac{\phi_u}{\phi_u + \phi_d} \pi_{0,H}, \quad B = \frac{\phi_u}{\phi_u + \phi_d} \frac{K_H^2}{c}. \quad (21)$$

Hence the firm's profit is

$$\bar{\Pi}_{LH}(\alpha; h) = (h - \alpha)(A + B\alpha). \quad (22)$$

Again (22) is a strictly concave quadratic in α , with unconstrained maximizer

$$\alpha_{LH}^\circ(h) = \frac{h}{2} - \frac{A}{2B}. \quad (23)$$

The Region R2 maximization is taken over the feasible interval $[\underline{\alpha}, \bar{\alpha}]$. Let $\alpha_{LH}^\circ(h)$ denote the unconstrained maximizer. The constrained solution is therefore

$$\alpha^*(h) = \min\{\bar{\alpha}, \max\{\underline{\alpha}, \alpha_{LH}^\circ(h)\}\}.$$

Equivalently, $\alpha^*(h) = \alpha_{LH}^\circ(h)$ when the interior solution lies in $[\underline{\alpha}, \bar{\alpha}]$, $\alpha^*(h) = \underline{\alpha}$ when $\alpha_{LH}^\circ(h) < \underline{\alpha}$, and $\alpha^*(h) = \bar{\alpha}$ when $\alpha_{LH}^\circ(h) > \bar{\alpha}$. Substituting into (20)–(22) gives

$$\bar{\pi}^* = A + B\alpha^*, \quad \bar{\Pi}^* = (h - \alpha^*)(A + B\alpha^*). \quad (24)$$

Region R3: all expert, full effort. If $\alpha \geq \bar{\alpha}$, then both types exert full effort. Promotions occur and demotions disappear, so starting from $s_1 = L$ the chain absorbs at H . Thus steady-state success probability is

$$\bar{\pi}_{HH} = \pi_{0,H} + K_H, \quad (25)$$

and firm profit is

$$\bar{\Pi}_{HH}(\alpha; h) = (h - \alpha)(\pi_{0,H} + K_H). \quad (26)$$

Since (26) is strictly decreasing in α , the firm chooses the smallest feasible bonus in Region R3, namely

$$\alpha^* = \bar{\alpha}. \quad (27)$$

Substituting (27) into (25)–(26) yields

$$\bar{\pi}^* = \pi_{0,H} + K_H, \quad \bar{\Pi}^* = (h - \bar{\alpha})(\pi_{0,H} + K_H). \quad (28)$$

Step 3: Comparison across regions and determination of h_2, h_3 . Let

$$\bar{\Pi}_{LL}^*(h), \quad \bar{\Pi}_{LH}^*(h), \quad \bar{\Pi}_{HH}^*(h)$$

denote the optimized values of (15), (22), and (26) within Regions R1, R2, and R3, respectively. The firm's problem is therefore

$$\bar{\Pi}^*(h) = \max\{\bar{\Pi}_{LL}^*(h), \bar{\Pi}_{LH}^*(h), \bar{\Pi}_{HH}^*(h)\}. \quad (29)$$

By (15) and (22), both $\bar{\Pi}_{LL}^*(h)$ and $\bar{\Pi}_{LH}^*(h)$ are continuous in h . Since Region R1 is optimal for all sufficiently small $h > 0$ (by (17)), define

$$h_2 = \inf\{h \geq h_1 : \bar{\Pi}_{LH}^*(h) \geq \bar{\Pi}_{LL}^*(h)\}. \quad (30)$$

By construction, for any $h_1 < h < h_2$ we have $\bar{\Pi}_{LH}^*(h) < \bar{\Pi}_{LL}^*(h)$. Moreover, throughout this interval the all-expert regime remains dominated, $\bar{\Pi}_{HH}^*(h) < \bar{\Pi}_{LL}^*(h)$. Hence the all-non-expert regime delivers the highest profit among the feasible regimes, so that

$$h_1 < h < h_2 \Rightarrow \bar{\Pi}^*(h) = \bar{\Pi}_{LL}^*(h).$$

In this region the firm therefore chooses the Region R1 interior solution

$$\alpha^* = \frac{h}{2} - \frac{c\pi_{0,L}}{2K_L^2} < \underline{\alpha},$$

which corresponds to the spot-checking regime described in part (iii) of the theorem.

Next, compare Regions R2 and R3. Since (22) and (26) are continuous in h , define

$$h_3 = \inf\{h \geq h_2 : \bar{\Pi}_{HH}^*(h) \geq \bar{\Pi}_{LH}^*(h)\}. \quad (31)$$

Then:

$$h_2 < h < h_3 \Rightarrow \bar{\Pi}^*(h) = \bar{\Pi}_{LH}^*(h),$$

and the firm chooses the mixed-regime bonus from (23), or $\alpha^* = \underline{\alpha}$ if the interior maximizer falls below $\underline{\alpha}$. This yields part (ii) of the theorem. Likewise,

$$h > h_3 \Rightarrow \bar{\Pi}^*(h) = \bar{\Pi}_{HH}^*(h),$$

and the firm chooses $\alpha^* = \bar{\alpha}$, yielding part (i).

Collecting (17), (30), and (31) gives

$$0 < h_1 \leq h_2 \leq h_3,$$

and the four regimes stated in the theorem: minimum effort for $h < h_1$, all- L spot-checks for $h_1 < h < h_2$, the mixed regime for $h_2 < h < h_3$, and all- H full diligence for $h > h_3$.

Final step: comparison with the static full-effort cutoff. Let

$$\bar{h}_L = \frac{c\pi_{0,L}}{K_L^2} + \frac{2c}{K_L} \quad (32)$$

denote the full-effort cutoff from Theorem 1 for a fixed L -type worker. At $h = \bar{h}_L$, the static firm is exactly indifferent between inducing interior effort and full effort from the L -type:

$$\max_{\alpha < c/K_L} (h - \alpha) \left(\pi_{0,L} + \frac{K_L^2}{c} \alpha \right) = \left(h - \frac{c}{K_L} \right) (\pi_{0,L} + K_L).$$

We claim that

$$\bar{\alpha} \leq \frac{c}{K_s}, \quad (33)$$

where $s \in \{L, H\}$ denotes the skill type whose incentive constraint binds at $\bar{\alpha}$. At the bonus level c/K_s , a worker of type s is already weakly willing to exert full effort in the corresponding static problem. In the dynamic environment, however, exerting full effort generates additional continuation benefits. A low-skill worker who exerts full effort increases the probability of promotion to H , while a high-skill worker who exerts full effort avoids the continuation-value loss associated with possible demotion. These dynamic considerations strictly strengthen the incentive to exert effort. Therefore, the bonus required to sustain full diligence for the binding type in the dynamic model cannot exceed the corresponding static threshold c/K_s , establishing (33).

Using (33) and (25), the full-diligence profit at $h = \bar{h}_L$ satisfies

$$\bar{\Pi}_{HH}^*(\bar{h}_L) = (\bar{h}_L - \bar{\alpha})(\pi_{0,H} + K_H) \geq \left(\bar{h}_L - \frac{c}{K_L} \right) (\pi_{0,H} + K_H).$$

Moreover, $\pi_{0,H} + K_H \geq \pi_{0,L} + K_L$, with strict inequality generically, because success probability is increasing in skill for any fixed effort. Hence

$$\bar{\Pi}_{HH}^*(\bar{h}_L) \geq \left(\bar{h}_L - \frac{c}{K_L} \right) (\pi_{0,L} + K_L),$$

again strictly generically. The right-hand side is exactly the static full-effort profit of the L -type at its own cutoff (32), and by construction this equals the best all- L interior benchmark at $h = \bar{h}_L$. Therefore, by the definition of h_3 in (31), the crossing from Region R2 to Region R3 occurs no later than $\bar{h}_L$, $h_3 \leq \bar{h}_L$, with strict inequality generically. $\square$

Proof of Theorem 3. The proof proceeds in two steps. First we show that under $\tau > 1/2$ and $\phi_u \leq \phi_d$ the bonus needed to pin non-experts exceeds the bonus needed to pin experts, i.e. $\alpha_L^* > \alpha_H^*$. Second, this implies that the mixed regime of Theorem 2(ii) cannot occur, so $h_2 = h_3$.

Step 1: Ordering of incentive thresholds. Let

$$u_s^\circ(\alpha) = \alpha\pi_{0,s} + \frac{\alpha^2 K_s^2}{2c}, \quad u_s^f(\alpha) = \alpha(\pi_{0,s} + K_s) - \frac{c}{2}$$

denote the worker's stage payoff under interior and full effort. Let α_H^* be the smallest bonus at which an expert prefers full effort to interior effort when the non-expert exerts interior effort. Standard Bellman algebra yields the expert indifference condition

$$u_H^\circ(\alpha_H^*) - u_H^f(\alpha_H^*) = \frac{\delta\phi_d}{1 - \delta(1 - \phi_d)} (u_H^\circ(\alpha_H^*) - u_L^\circ(\alpha_H^*)). \quad (34)$$

Now compare static losses from deviating from interior effort. Since $\tau > 1/2$ we have

$$K_H - K_L = (1 - p)q\theta(2\tau - 1) > 0,$$

so $K_H > K_L$. For any $\alpha < c/K_L$,

$$u_s^\circ(\alpha) - u_s^f(\alpha) = \frac{(c - K_s\alpha)^2}{2c}.$$

Because $K_H > K_L$,

$$u_L^\circ(\alpha) - u_L^f(\alpha) > u_H^\circ(\alpha) - u_H^f(\alpha). \quad (35)$$

Evaluating (35) at $\alpha = \alpha_H^*$ and using (34) gives

$$u_L^\circ(\alpha_H^*) - u_L^f(\alpha_H^*) > \frac{\delta\phi_d}{1 - \delta(1 - \phi_d)} (u_H^\circ(\alpha_H^*) - u_L^\circ(\alpha_H^*)).$$

Since $\phi_u \leq \phi_d$, the continuation benefit from promotion for the low-skill worker is weakly smaller than the demotion-loss benefit for the expert. Hence at $\alpha = \alpha_H^*$ the low-skill worker still prefers interior effort, implying

$$\alpha_L^* > \alpha_H^*.$$

Step 2: Disappearance of the mixed regime. Because $\alpha_L^* > \alpha_H^*$, any bonus high enough to induce full effort from non-experts also induces full effort from experts. Hence there is no bonus interval on which

$$x_L^*(\alpha) = 1 \quad \text{and} \quad x_H^*(\alpha) \in (0, 1).$$

Therefore the mixed regime of Theorem 2(ii) cannot arise.

Consequently the firm's steady-state choice collapses to two possibilities: either the worker remains permanently non-expert (if $\alpha < \alpha_L^*$), or the chain absorbs at the expert state (if $\alpha \geq \alpha_L^*$). The cutoff between these regions defines $h_2 = h_3$, and in steady state $\mu_H^* \in \{0, 1\}$. $\square$

Proof of Proposition 6. In the high-complexity regime ($\tau > 1/2$ and $\phi_u \leq \phi_d$), Theorem 3 implies that the mixed steady state of Theorem 2(ii) disappears. Thus under any stationary bonus α the long-run outcome is one of two possibilities:

- (i) $\alpha < \alpha_L^*$: the low-skill worker exerts interior effort, promotions never occur, and the chain remains permanently in state L ;
- (ii) $\alpha \geq \alpha_L^*$: both types exert full effort and the chain absorbs in H .

Hence the firm's steady-state outcome depends only on whether α lies below or above the non-expert pinning threshold α_L^* .

All- L region. If $\alpha < \alpha_L^*$, both types exert interior effort and the steady state remains at L . The steady-state success probability and profit are therefore

$$\bar{\pi}(\alpha) = \pi_{0,L} + \frac{K_L^2}{c}\alpha, \quad \bar{\Pi}(\alpha) = (h - \alpha)\bar{\pi}(\alpha). \quad (36)$$

Expression (36) is strictly concave in α with unique maximizer

$$\alpha_{LL}^\circ(h) = \frac{h}{2} - \frac{c\pi_{0,L}}{2K_L^2}. \quad (37)$$

Importantly, $\alpha_{LL}^\circ(h)$ depends only on $(h, \pi_{0,L}, K_L, c)$ and is therefore independent of (ϕ_u, ϕ_d) . Thus neither the optimal bonus nor the steady-state outcome depends on learning or skill decay in this region.

Pinned region. If $\alpha \geq \alpha_L^*$, both types exert full effort and the chain absorbs in H , yielding

$$\bar{\pi}^* = \pi_{0,H} + K_H, \quad \bar{\Pi}^*(h) = (h - \alpha_L^*)(\pi_{0,H} + K_H). \quad (38)$$

The pinning threshold α_L^* is determined by the indifference condition between full and interior effort for a non-expert,

$$G_L(\alpha) \equiv u_L^\circ(\alpha) - \left[u_L^{\text{full}}(\alpha) + \delta\phi_u \frac{u_H^{\text{full}}(\alpha) - u_L^{\text{full}}(\alpha)}{1 - \delta(1 - \phi_u)} \right] = 0. \quad (39)$$

Comparative statics of α_L^* . Differentiating the root condition $G_L(\alpha_L^*) = 0$ and applying the implicit function theorem gives

$$\frac{\partial \alpha_L^*}{\partial \phi_u} = -\frac{\partial G_L / \partial \phi_u}{\partial G_L / \partial \alpha}, \quad \frac{\partial \alpha_L^*}{\partial \phi_d} = -\frac{\partial G_L / \partial \phi_d}{\partial G_L / \partial \alpha}.$$

At the first root of G_L , we have $\partial G_L / \partial \alpha < 0$. Moreover, the continuation benefit from promotion implies

$$\frac{\partial G_L}{\partial \phi_u} < 0, \quad \frac{\partial G_L}{\partial \phi_d} = 0.$$

Therefore

$$\frac{\partial \alpha_L^*}{\partial \phi_u} < 0, \quad \frac{\partial \alpha_L^*}{\partial \phi_d} = 0. \quad (40)$$

Implications for firm outcomes. From (38), steady-state success does not depend on learning or skill decay,

$$\frac{\partial \bar{\pi}^*}{\partial \phi_u} = \frac{\partial \bar{\pi}^*}{\partial \phi_d} = 0.$$

However, profit varies through the incentive cost α_L^* . Using (40),

$$\frac{\partial \bar{\Pi}^*}{\partial \phi_u} > 0, \quad \frac{\partial \bar{\Pi}^*}{\partial \phi_d} = 0.$$

Finally, for small h the interior optimum $\alpha_{LL}^\circ(h)$ lies below α_L^* and does not depend on (ϕ_u, ϕ_d) . As ϕ_u rises the threshold α_L^* falls, shifting the cutoff at which the firm switches to full diligence. Once in the pinned region, the optimal bonus decreases and profit rises. Changes in ϕ_d leave both the threshold and the optimal bonus unchanged.

Thus the optimal bonus is weakly decreasing in learning and independent of skill decay, while steady-state success and profit are weakly increasing in learning and independent of skill decay. $\square$

Proof of Theorem 4. The proof proceeds in two steps. First, we show that under the low-complexity assumptions $\tau < 1/2$ and $\phi_u \geq \phi_d$, the worker incentive thresholds satisfy $\alpha_L^* < \alpha_H^*$ on an open set of primitives, so the mixed regime is feasible. Second, we show that on an open set of such primitives the firm strictly prefers that mixed regime for an open interval of output values h , implying $h_2 < h_3$.

Step 1: The mixed regime is generically feasible. Under $\tau < 1/2$,

$$K_L - K_H = (1 - p)q\theta(1 - 2\tau) > 0,$$

so low-skill workers have the larger marginal return to effort. This is the key low-complexity feature: effort is more productive for non-experts than for experts.

Let α_L^* denote the smallest bonus that makes a non-expert indifferent between interior effort and full effort when experts exert interior effort, and let α_H^* denote the smallest bonus that makes an expert indifferent between interior effort and full effort once non-experts are pinned. These are characterized by the quadratic incentive conditions already derived in the appendix.

For the non-expert threshold, the indifference condition takes the form

$$a_L \alpha^2 - b_L \alpha + \frac{c}{2} = 0, \quad (41)$$

where

$$a_L > 0, \quad b_L = \frac{(1 - \delta)K_L + \delta\phi_u K_H + \delta\phi_u(\pi_{0,H} - \pi_{0,L})}{1 - \delta(1 - \phi_u)}.$$

Hence α_L^* is the smaller root of (41).

For the expert threshold, conditional on non-experts being pinned, the indifference condition takes the form

$$a_H \alpha^2 - b_H \alpha + k_H = 0, \quad (42)$$

where

$$a_H > 0, \quad b_H = \frac{\pi_{0,H} + K_H}{1 - \delta} - \frac{(1 - \delta(1 - \phi_u))\pi_{0,H} + \delta\phi_d(\pi_{0,L} + K_L)}{(1 - \delta(1 - \phi_u))(1 - \delta(1 - \phi_d)) - \delta^2\phi_u\phi_d},$$

and $k_H > 0$ is the corresponding constant term from the appendix expression. Thus α_H^* is the smaller root of (42).

Because the coefficients in (41)–(42) depend continuously on the primitives, the thresholds α_L^* and α_H^* are continuous wherever the corresponding roots are well defined and distinct.

We now show that $\alpha_L^* < \alpha_H^*$ holds on a nonempty open set. Consider the benchmark case $\phi_u = \phi_d = 0$. Then continuation values are irrelevant, so the thresholds reduce to the static full-effort thresholds:

$$\alpha_L^* = \frac{c}{K_L}, \quad \alpha_H^* = \frac{c}{K_H}.$$

Since $K_L > K_H$, we have

$$\frac{c}{K_L} < \frac{c}{K_H},$$

hence

$$\alpha_L^* < \alpha_H^*.$$

By continuity of the roots, this strict inequality persists for all (ϕ_u, ϕ_d) in some open neighborhood of $(0, 0)$ satisfying $\phi_u \geq \phi_d$. Therefore, on a nonempty open set of low-complexity primitives,

$$\alpha_L^* < \alpha_H^*. \quad (43)$$

Whenever (43) holds, any bonus $\alpha \in [\alpha_L^*, \alpha_H^*)$ induces

$$x_L^*(\alpha) = 1, \quad x_H^*(\alpha) = \frac{\alpha K_H}{c} \in (0, 1),$$

so the induced skill process is a two-state birth–death chain with stationary distribution

$$\mu_L = \frac{\phi_d}{\phi_u + \phi_d}, \quad \mu_H = \frac{\phi_u}{\phi_u + \phi_d}. \quad (44)$$

Hence the mixed regime is feasible and, for such bonuses,

$$\mu_H^* = \frac{\phi_u}{\phi_u + \phi_d} \in (0, 1).$$

Step 2: The mixed regime is generically optimal for some h . On the three relevant regions from Theorem 2, the firm's steady-state profit is

$$\Pi_{LL}(\alpha; h) = (h - \alpha) \left(\pi_{0,L} + \frac{K_L^2}{c} \alpha \right), \quad \alpha < \alpha_L^*, \quad (45)$$

$$\Pi_{LH}(\alpha; h) = (h - \alpha)(A + B\alpha), \quad \alpha \in [\alpha_L^*, \alpha_H^*], \quad (46)$$

$$\Pi_{HH}(\alpha; h) = (h - \alpha)(\pi_{0,H} + K_H), \quad \alpha \geq \alpha_H^*, \quad (47)$$

where

$$A = \frac{\phi_d}{\phi_u + \phi_d}(\pi_{0,L} + K_L) + \frac{\phi_u}{\phi_u + \phi_d}\pi_{0,H}, \quad B = \frac{\phi_u}{\phi_u + \phi_d} \frac{K_H^2}{c}.$$

The first two are strictly concave quadratics in α , and the third is strictly decreasing in α . Their constrained optimizers are therefore

$$\alpha_{LL}^\circ(h) = \frac{h}{2} - \frac{c\pi_{0,L}}{2K_L^2}, \quad \alpha_{LH}^\circ(h) = \frac{h}{2} - \frac{A}{2B}, \quad \alpha_{HH}^* = \alpha_H^*.$$

Because of (43), the interval (α_L^*, α_H^*) is nonempty. Hence there is a nonempty open interval of h such that

$$\alpha_{LH}^\circ(h) \in (\alpha_L^*, \alpha_H^*). \quad (48)$$

Indeed, $\alpha_{LH}^\circ(h)$ is affine and strictly increasing in h , so its preimage of the open interval (α_L^*, α_H^*) is itself an open interval.

Fix any h satisfying (48). At such an h , the mixed-region objective is optimized at an interior point, with value

$$\Pi_{LH}^*(h) = \left(\frac{h}{2} + \frac{A}{2B} \right) \left(\frac{A}{2} + \frac{Bh}{2} \right). \quad (49)$$

Similarly, the all- L interior optimum is

$$\Pi_{LL}^*(h) = \left(\frac{h}{2} + \frac{c\pi_{0,L}}{2K_L^2} \right) \left(\frac{\pi_{0,L}}{2} + \frac{hK_L^2}{2c} \right), \quad (50)$$

provided $\alpha_{LL}^\circ(h) < \alpha_L^*$, while the all- H full-effort branch is

$$\Pi_{HH}^*(h) = (h - \alpha_H^*)(\pi_{0,H} + K_H). \quad (51)$$

We now construct a strict witness. Take (ϕ_u, ϕ_d) sufficiently small with $\phi_u \geq \phi_d$, so that (43) holds and

$$\mu_H = \frac{\phi_u}{\phi_u + \phi_d} \in (0, 1).$$

Then A is a strict convex combination of $\pi_{0,L} + K_L$ and $\pi_{0,H}$, and $B > 0$. For intermediate values of h , the mixed branch (49) strictly improves on the all- L branch because it preserves the large marginal return to effort of low-skill workers while also generating a positive expert share.

At the same time, for such intermediate h , the all- H branch (51) is still dominated because inducing full effort from experts is too expensive relative to the value of the additional output. Hence there exists at least one strict witness parameter vector and one open interval of h for which

$$\Pi_{LH}^*(h) > \max\{\Pi_{LL}^*(h), \Pi_{HH}^*(h)\}.$$

Because all thresholds and optimized profit branches depend continuously on the primitives, this

strict inequality persists on an open neighborhood of that witness. Therefore, on a nonempty open set of low-complexity primitives, there exists an open interval of output values for which the mixed regime is strictly optimal. Equivalently, $h_2 < h_3$; hence, region (ii) of Theorem 2 generically exists. For all $h \in (h_2, h_3)$, the steady-state expert share is given by (44), so in particular $\mu_H^* \in (0, 1)$. $\square$

Proof of Proposition 7. By Theorem 4, in the low-complexity regime there exist cutoffs $h_2 < h_3$ such that the optimal steady-state regime is: LL if $h < h_2$, LH if $h_2 < h < h_3$, and HH if $h > h_3$. Accordingly,

$$\mu_H^*(h) = \begin{cases} 0, & h < h_2, \\ \frac{\phi_u}{\phi_u + \phi_d}, & h_2 < h < h_3, \\ 1, & h > h_3. \end{cases} \quad (52)$$

Thus the only way μ_H^* can vary with (ϕ_u, ϕ_d) is through the mixed-region share $\phi_u/(\phi_u + \phi_d)$ and through movements in the boundary h_3 separating LH from HH.

We first record two comparative statics that follow from the worker incentive constraints in the low-complexity regime. The expert-pinning threshold α_H^* is strictly decreasing in ϕ_d and strictly increasing in ϕ_u : when demotion is more severe, experts are easier to pin at full effort, whereas when return to expertise is faster, the loss from demotion is smaller and experts are harder to pin. In the mixed LH regime, however, the continuation value of promotion depends on the value of the high-skill state, and that value varies with ϕ_d . Hence changes in ϕ_d affect equilibrium continuation values and, through them, the equilibrium pinning threshold α_L^* .

These forces imply the following monotonicity of the upper regime cutoff. Fix all primitives other than ϕ_d . As ϕ_d rises, the HH region becomes more attractive (because α_H^* falls) while the LH region becomes less attractive (because μ_H^{LH} falls). Hence, generically,

$$h_3 \text{ is strictly decreasing in } \phi_d. \quad (53)$$

Similarly, fixing all primitives other than ϕ_u , a rise in ϕ_u makes HH less attractive (because α_H^* rises) and LH more attractive (because μ_H^{LH} rises), so generically

$$h_3 \text{ is strictly increasing in } \phi_u. \quad (54)$$

We now prove the claimed non-monotonicities.

Variation in ϕ_d . Fix ϕ_u and choose primitives in the open set from Theorem 4 for which $h_2 < h_3$. By continuity of the cutoff functions, there exist $\underline{\phi}_d < \bar{\phi}_d$ and some h such that

$$h_2(\phi_d) < h < h_3(\underline{\phi}_d) \quad \text{and} \quad h > h_3(\bar{\phi}_d).$$

Thus the optimal regime is LH at $\underline{\phi}_d$ and HH at $\bar{\phi}_d$. By (52),

$$\mu_H^*(\underline{\phi}_d) = \frac{\phi_u}{\phi_u + \underline{\phi}_d} \in (0, 1), \quad \mu_H^*(\bar{\phi}_d) = 1.$$

Moreover, while the regime remains LH, the expert share $\phi_u/(\phi_u + \phi_d)$ is strictly decreasing in ϕ_d . Hence μ_H^* first decreases in ϕ_d and then jumps up to 1 when the firm switches from LH to HH. Therefore μ_H^* is generically non-monotone in ϕ_d .

Variation in ϕ_u . Fix ϕ_d and again choose primitives in the open set from Theorem 4. By continuity and (54), there exist $\underline{\phi}_u < \bar{\phi}_u$ and some h such that

$$h > h_3(\underline{\phi}_u) \quad \text{and} \quad h_2(\phi_u) < h < h_3(\bar{\phi}_u).$$

Thus the optimal regime is HH at $\underline{\phi}_u$ and LH at $\bar{\phi}_u$. Hence

$$\mu_H^*(\underline{\phi}_u) = 1, \quad \mu_H^*(\bar{\phi}_u) = \frac{\bar{\phi}_u}{\phi_u + \bar{\phi}_d} \in (0, 1).$$

Once the regime has switched to LH, the expert share $\phi_u/(\phi_u + \phi_d)$ is strictly increasing in ϕ_u . Therefore μ_H^* jumps down from 1 when the firm switches from HH to LH and then rises within the mixed regime, so μ_H^* is generically non-monotone in ϕ_u . $\square$

Proof of Proposition 8. Work in the low-complexity regime $\tau < 1/2$, so Theorem 4 applies. Fix a stake h in the intermediate range where, for some values of (ϕ_u, ϕ_d) , the regimes LH and HH are both feasible. Write

$$\mu \equiv \frac{\phi_u}{\phi_u + \phi_d} \in (0, 1), \quad \Theta_0 \equiv \pi_{0,L} + K_L, \quad \Theta_1 \equiv \frac{K_H^2}{c}.$$

In the LH regime, steady-state success is affine in the bonus:

$$\bar{\pi}^{LH}(\alpha; \mu) = K_0(\mu) + K_1(\mu)\alpha, \quad K_0(\mu) = \Theta_0 + \mu(\pi_{0,H} - \Theta_0), \quad K_1(\mu) = \Theta_1\mu, \quad (55)$$

and the firm solves

$$\bar{\Pi}^{LH}(\alpha; h, \mu) = (h - \alpha)(K_0(\mu) + K_1(\mu)\alpha) \quad \text{over } \alpha \in [\alpha_L^*, \alpha_H^*]. \quad (56)$$

In the HH regime,

$$\bar{\pi}^{HH} = \pi_{0,H} + K_H, \quad \bar{\Pi}^{HH}(h; \phi_u, \phi_d) = (h - \alpha_H^*(\phi_u, \phi_d))(\pi_{0,H} + K_H). \quad (57)$$

We use the following comparative statics for the low-complexity case in the LH regime:

$$\frac{\partial \alpha_L^*}{\partial \phi_u} < 0, \quad \frac{\partial \alpha_L^*}{\partial \phi_d} > 0, \quad \frac{\partial \alpha_H^*}{\partial \phi_u} > 0, \quad \frac{\partial \alpha_H^*}{\partial \phi_d} < 0. \quad (58)$$

Here the dependence of α_L^* on ϕ_d is indirect: although demotion risk does not enter the low-skill worker's current-period payoff directly, it affects continuation values through the value of reaching and remaining in the high-skill state. As a result, changes in ϕ_d alter the non-expert pinning threshold as well. Moreover, Proposition 7 shows that generically the optimal regime switches from HH to LH as ϕ_u rises, and from LH to HH as ϕ_d rises.

1. Non-monotonicity of α^* . When the LH optimum is interior, the first-order condition for (56) gives

$$\hat{\alpha}(h; \mu) = \frac{h}{2} - \frac{K_0(\mu)}{2K_1(\mu)}. \quad (59)$$

Since

$$\frac{K_0(\mu)}{K_1(\mu)} = \frac{\Theta_0}{\Theta_1} \frac{1}{\mu} + \frac{\pi_{0,H} - \Theta_0}{\Theta_1},$$

we have

$$\frac{\partial \hat{\alpha}}{\partial \mu} > 0. \quad (60)$$

Hence $\hat{\alpha}$ rises with ϕ_u and falls with ϕ_d .

Now consider the optimal bonus α^* . For variation in ϕ_u , Proposition 7 implies that generically the optimal regime switches from HH to LH at some $\hat{\phi}_u$. On the HH side, $\alpha^* = \alpha_H^*$ and (58) implies that α^* rises in ϕ_u . At the switch, the optimizer jumps down to the LH branch, which is either $\hat{\alpha}$ or α_L^* . Since $\alpha_H^* > \alpha_L^*$ generically in the mixed-feasible region, this creates a downward jump. Thus α^* is generically non-monotone in ϕ_u .

For variation in ϕ_d , the relevant branches move in opposite directions: inside LH, $\hat{\alpha}$ falls in ϕ_d by (60), whereas the boundary branch α_L^* rises in ϕ_d by (58); in HH, α_H^* falls in ϕ_d . Hence, whenever the optimum switches between the interior-LH branch and the boundary-LH branch, or between LH and HH, the slopes differ. Therefore α^* is generically non-monotone in ϕ_d as well.

2. Non-monotonicity of $\bar{\pi}^*$ and $\bar{\Pi}^*$. Suppose first that the LH optimum is interior, so $\alpha^* = \hat{\alpha}(h; \mu)$. Substituting (59) into (55) gives

$$\bar{\pi}^{LH,*}(h; \mu) = \frac{1}{2} (K_0(\mu) + hK_1(\mu)). \quad (61)$$

Hence

$$\frac{\partial \bar{\pi}^{LH,*}}{\partial \mu} = \frac{1}{2} [(\pi_{0,H} - \Theta_0) + h\Theta_1].$$

This is strictly positive on an open interval of intermediate stakes h . Likewise, substituting (59) into (56) yields

$$\bar{\Pi}^{LH,*}(h; \mu) = \frac{(K_0(\mu) + hK_1(\mu))^2}{4K_1(\mu)}, \quad (62)$$

and direct differentiation gives

$$\frac{\partial \bar{\Pi}^{LH,*}}{\partial \mu} = \frac{(\pi_{0,H} - \Theta_0 + h\Theta_1)^2 - (\Theta_0/\mu)^2}{4\Theta_1},$$

which is also strictly positive on an open interval of intermediate h . Thus, on an open set of mixed-regime stakes,

$$\bar{\pi}^{LH,*} \text{ and } \bar{\Pi}^{LH,*} \text{ increase in } \phi_u \text{ and decrease in } \phi_d. \quad (63)$$

In the HH regime, by (57),

$$\bar{\pi}^{HH} = \pi_{0,H} + K_H$$

is constant in (ϕ_u, ϕ_d) , while

$$\bar{\Pi}^{HH}(h; \phi_u, \phi_d) = (h - \alpha_H^*)(\pi_{0,H} + K_H)$$

decreases in ϕ_u and increases in ϕ_d .

Now combine these within-regime monotonicities with Proposition 7. Generically, as ϕ_u rises the optimal regime switches from HH to LH. Across that switch, $\bar{\Pi}^*$ moves from a branch that decreases in ϕ_u to a branch that increases in ϕ_u , so $\bar{\Pi}^*$ is generically non-monotone in ϕ_u . Likewise, $\bar{\pi}^*$ is constant in HH but increasing in LH on the open set identified above, with a discrete downward change at the HH→LH switch; hence $\bar{\pi}^*$ is generically non-monotone in ϕ_u .

Similarly, as ϕ_d rises, Proposition 7 gives a generic LH→HH switch. By (63), both $\bar{\pi}^{LH,*}$ and $\bar{\Pi}^{LH,*}$ decrease in ϕ_d on an open set of mixed-regime stakes, whereas in HH, $\bar{\pi}^{HH}$ is constant and $\bar{\Pi}^{HH}$ increases in ϕ_d . Therefore both $\bar{\pi}^*$ and $\bar{\Pi}^*$ are generically non-monotone in ϕ_d as well. $\square$

B Baseline Parameters

In a study of computer-assisted electrocardiogram (ECG) interpretation, [Kashou et al. \(2023\)](#) enlisted a diverse group of health care providers to interpret ECGs with both urgent and nonurgent findings. In the first of two phases of the study, participants received the ECGs accompanied by a clinical statement about the patient. In the second phase participants were also provided with a computerized ECG interpretation (CEI). For each phase, the study collected information about accuracy (percentage of correctly identified ECG findings) and the time spent on each interpretation. The study concluded that across all providers, use of the CEI increased both accuracy and efficiency.

Here we use the results from the second phase of the study as data, to see how the CEI - a relatively simple type of AI - interacts with human effort across a range of skill levels. This study is particularly useful for our purposes because participants had a wide range of expertise, including non-physicians (nurses, advanced practice providers, medical students) and physicians (primary care, residents, cardiology fellows-in-training). To derive the baseline parameters of our model, we will focus on results from the 137 medical students and the 123 cardiology fellows-in-training in the study. The former group had the lowest level of accuracy in phase 1, so we classify them as the least-capable *non-experts* of all ECG readers in the study, with skill parameter $s = 0$. The cardiology fellows-in-training had the highest level of accuracy, so they are the most-capable *experts* with $s = 1$. Given the accuracy and time required to read each ECG for these two groups, we will infer the model parameter values p, τ, q and θ . At the end of this section we will also consider the cost and value parameters, as well as the parameters for the dynamic model.

	Interpretation Accuracy		Interpretation Time	
	mean	std dev	mean (sec.)	std dev (sec.)
	$\hat{\pi}_s$	$\hat{\sigma}_s^\pi$	$\hat{t}_s$	$\hat{\sigma}_s^t$
Medical students ($s = 0$)	0.55	0.23	92	25
Cardiology fellows-in-training ($s = 1$)	0.74	0.16	96	31

Table 1. ECG reading accuracy and interpretation times given CEI; numbers derived from Supplemental Table 1 of [Kashou et al. \(2023\)](#)

Table 1 shows the observed accuracy and interpretation times for the two groups of subjects, as well as the standard deviation of observations for each subject from each group. We will convert the interpretation time t into our model’s measure of effort x , under the assumption that a longer interpretation time implies more effort and a larger probability of an accurate interpretation.

Let t be the average interpretation time of a subject during the experiment. To bound effort, we assume that $t \in [\hat{t}_s - 2\hat{\sigma}_s^t, \hat{t}_s + 2\hat{\sigma}_s^t]$, where the lower bound represents minimum effort and the upper bound is maximum effort. Define $t' = t - (\hat{t}_s - 2\hat{\sigma}_s^t)$ as the subject’s time-effort above the minimum. We assume a linear relationship between this quantity and accuracy, given the CEI:

$$\pi_s = \pi_{0,s} + \beta_s t' + \epsilon$$

where π_s is the accuracy of an experimental subject from skill group s , $\pi_{0,s}$ is the subject’s expected

accuracy at minimum effort, and ϵ is an error term. Because the actual outcome of each reading is binary - accurate or not accurate - we could instead use a probit or logit model. For the sake of simplicity we will use this linear model, and given the data above, we find reasonable parameter values that produce predicted probabilities between 0 and 1.

The parameter β_s is the expected increase in accuracy for each additional second of effort for a subject from skill group s . Instead of assuming values of β_s , it is more intuitive to assume a level of *correlation* between interpretation time and accuracy. Let $\hat{r}_{\pi,t}$ be the sample correlation coefficient between π_s and t' , assumed to be the same for both s , and therefore the least-squares estimate for β_s is $\beta_s^* = (\hat{\sigma}_s^\pi / \hat{\sigma}_s^t) \hat{r}_{\pi,t}$. [Kashou et al. \(2023\)](#) did not report a sample correlation coefficient between time and accuracy, but data from [Monteiro et al. \(2024\)](#) (Table 2 and Figure 6) indicates a significant positive correlation, although there is a wide possible range of possible correlation values from their data. For our baseline parameters we will assume that the correlation $\hat{r}_{\pi,t} = 0.25$, although we will examine the impact of variations from this baseline in the main text.

To match our original model, we must also convert the observed effort t' to an effort level $x_t \in [0, 1]$. Define $x = t' / 4\hat{\sigma}_s^t$ and therefore we can rewrite the linear model above as,

$$\pi_s = \pi_{0,s} + 4\hat{\sigma}_s^t \beta_s x + \epsilon.$$

Define $K_s = 4\hat{\sigma}_s^t \beta_s$, and we recover our original model $\pi(s, x) = \pi_{0,s} + K_s x$, with $\pi_{0,s} = 1 - (1 - p)[(1 - \tau)(1 - \theta s) + \tau]$ and $K_s = (1 - p)q[(1 - \tau)(1 - \theta s) + \tau\theta s]$.

Finally, define $\hat{K}_s = 4\hat{\sigma}_s^t \beta_s^*$ and the slopes $\hat{K}_s$ can be calculated from the data in Table 1. Substituting $s = 0$ for the medical students and $s = 1$ for the cardiology fellows-in-training, we have:

$$\begin{aligned}\hat{K}_L &= (1 - p)q(1 - \tau) \\ \hat{K}_H &= (1 - p)q[(1 - \tau)(1 - \theta) + \tau\theta] \\ \hat{\pi}_L &= p + \frac{1}{2}\hat{K}_L \\ \hat{\pi}_H &= [1 - (1 - p)[(1 - \tau)(1 - \theta) + \tau]] + \frac{1}{2}\hat{K}_H.\end{aligned}$$

This is four equations with four unknowns: p, τ, q and θ . After some algebra, we find that there are two sets of solutions, determined by the solutions to a quadratic equation. Calculating from the values in Table 1, one of the solution sets has a value $\tau < 0$, while the other solution set is consistent with our model, with values $p = 0.44, \tau = 0.32, q = 0.59$, and $\theta = 0.58$. These are our baseline parameters, and figure 8 shows, for these parameters, the probability of a correct result given effort x . Note that with $\tau = 0.32 < 0.5$, skill and effort are substitutes, and the slope of the highest-skilled expert is shallower than the slope for the lowest-skilled worker.

Now we propose values for the cost parameter c and the value of a correct answer h . For this example, from Table 1, the difference $4\hat{\sigma}_s^t$ between maximum and minimum effort is approximately two minutes. According to the placement service Indeed, the annual wage for a cardiology fellowship is \$233,363 ([Smith, 2025](#)). With maximum duty hours of 80 hours/week ([Dartmouth Hitchcock Medical Center, 2026](#)), the wage is \$58.34/hour or \$0.97/minute. Therefore, the maximum cost of effort with

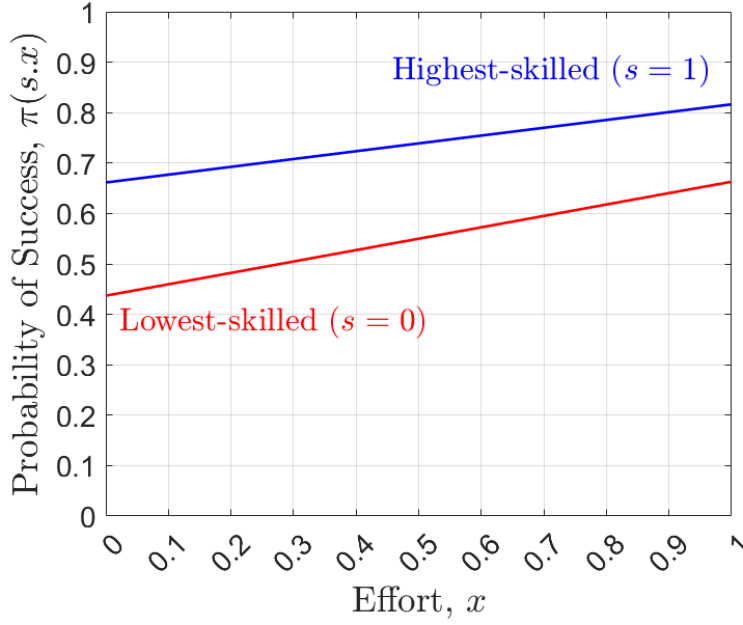


Figure 8. Probability of correct result, given effort x and baseline parameters

$x = 1$ is $(1/2)cx^2 = (1/2)c = (2 \text{ minutes})(\$0.97/\text{minute})$, so we set $c = 4 * 0.97 = 3.89$. Also assume that the value of a correct result h is 10X the maximum effort, 38.9. This may be too low or high for certain outcomes, and we will vary h in the main body of the paper, e.g., Figures 3 and 5.

Finally, for the dynamic model, we need to define the levels of expertise and the learning/decay rates for two skill levels: “non-experts” and “experts.” In the analysis above we assumed that medical students have the very lowest level of expertise, $s = 0$, so we assume that our non-expert in the dynamic model is superior to the medical student but not by much, with $s = 0.2$. We assume that the cardiology fellow-in-training has the very highest level, $s = 1$, and we set our expert as a bit inferior with $s = 0.8$. For expertise learning and decay over time, assume that the time-period is one year, with discount factor $\delta = 0.95$, corresponding to an annual discount rate of 5.3% on cash flows. We assume that over the course of the year a non-expert at full effort has a 10% chance of becoming an expert, $\phi_u = 0.1$. As for ϕ_d , a lack of attention will eventually lead to a decay in expertise. ECG interpretation, however, is a relatively mature field, so it is not necessary to put in effort to ensure that skills are up-to-date. Therefore we assume that the skill decay rate is less than the learning rate, with $\phi_d = 0.05$.